7 Space

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 7

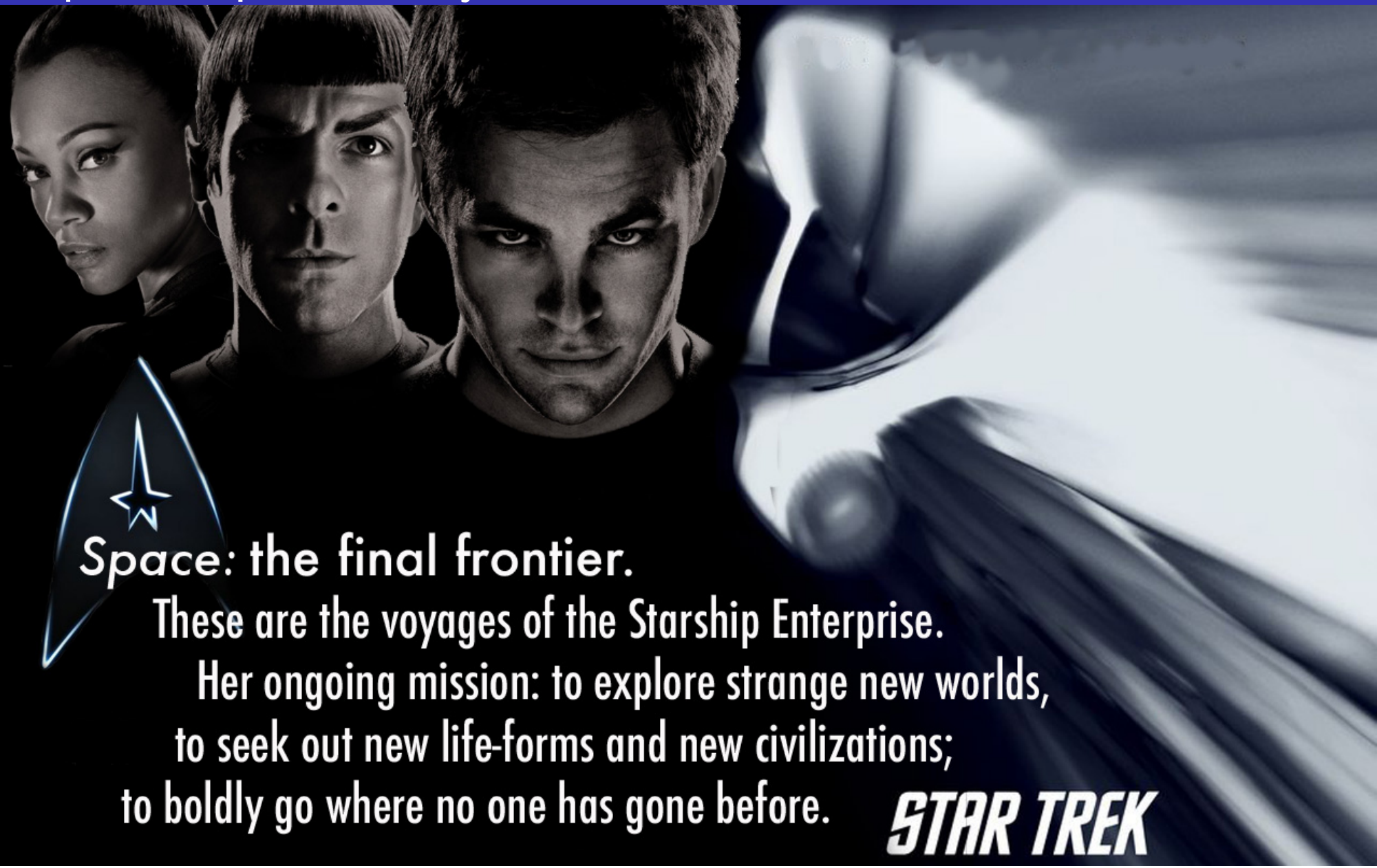
Space

Tuesday 27 October 2026

Announcements

- **Midterm test:**
  - *Date:* Tuesday 24 November 2026 (tentative)
  - *Time available:* 2:30–5:20 p.m. (test duration: 90 minutes)
  - *Location:* in class, ETB 227
  - Test structure will be discussed in the class before the test.
- **Assignment 3** is due Thursday 5 November 2026 at 11:59 p.m.
- **Assignment 4** is due Thursday 19 November 2026 at 11:59 p.m.
  - Make sure *you personally* can do the question on calculating $\mathcal{R}_0$ on this assignment *before* the midterm test.

Spatial Epidemic Dynamics



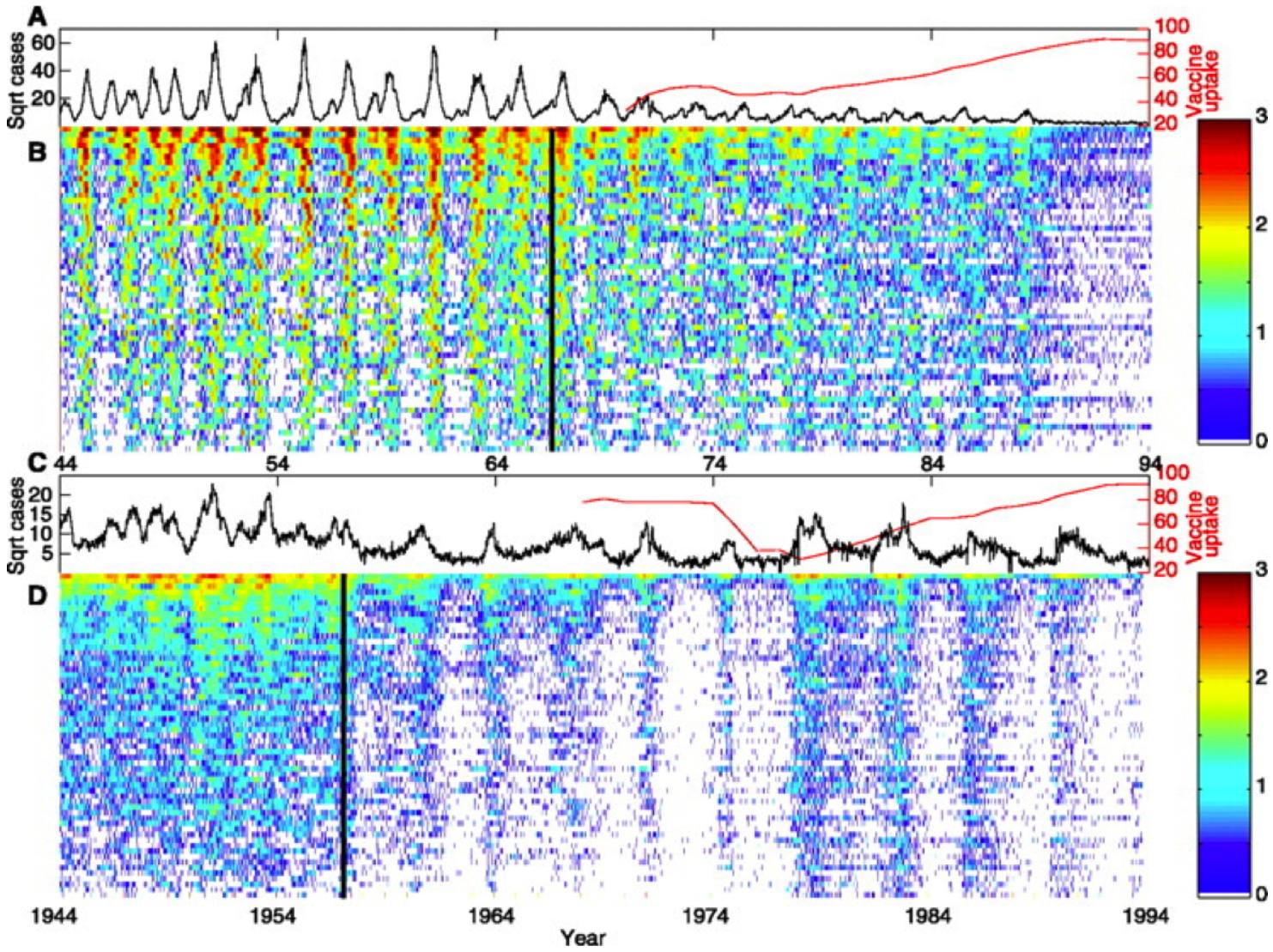
Something to think about

- All of our analysis has been of temporal patterns of epidemics
- What about spatial patterns?
- What problems are suggested by observed spatial epidemic patterns?
- Can spatial epidemic data suggest improved strategies for control?
- Can we reduce the eradication threshold below $p_{\mathrm{crit}} = 1 - \frac{1}{\mathcal{R}_0}$?

Measles and Whooping Cough in 60 UK cities

Measles

Whooping Cough



Rohani, Earn & Grenfell (1999) *Science* **286**, 968–971

Better Control? Eradication?

- The term-time forced SEIR model successfully predicts past patterns of epidemics of childhood diseases
- Can we manipulate epidemics predictably so as to increase probability of eradication?
- Can we eradicate measles?

Idea for eradicating measles

- Try to re-synchronize measles epidemics in the UK and, moreover, synchronize measles epidemics worldwide: synchrony is good
- Devise new vaccination strategy that tends to synchronize...
- Avoid spatially structured epidemics...
- Time to think about the mathematics of synchrony...
- But analytical theory of synchrony in a periodically forced system of differential equations is mathematically demanding...
- So let's consider a much simpler biological model...

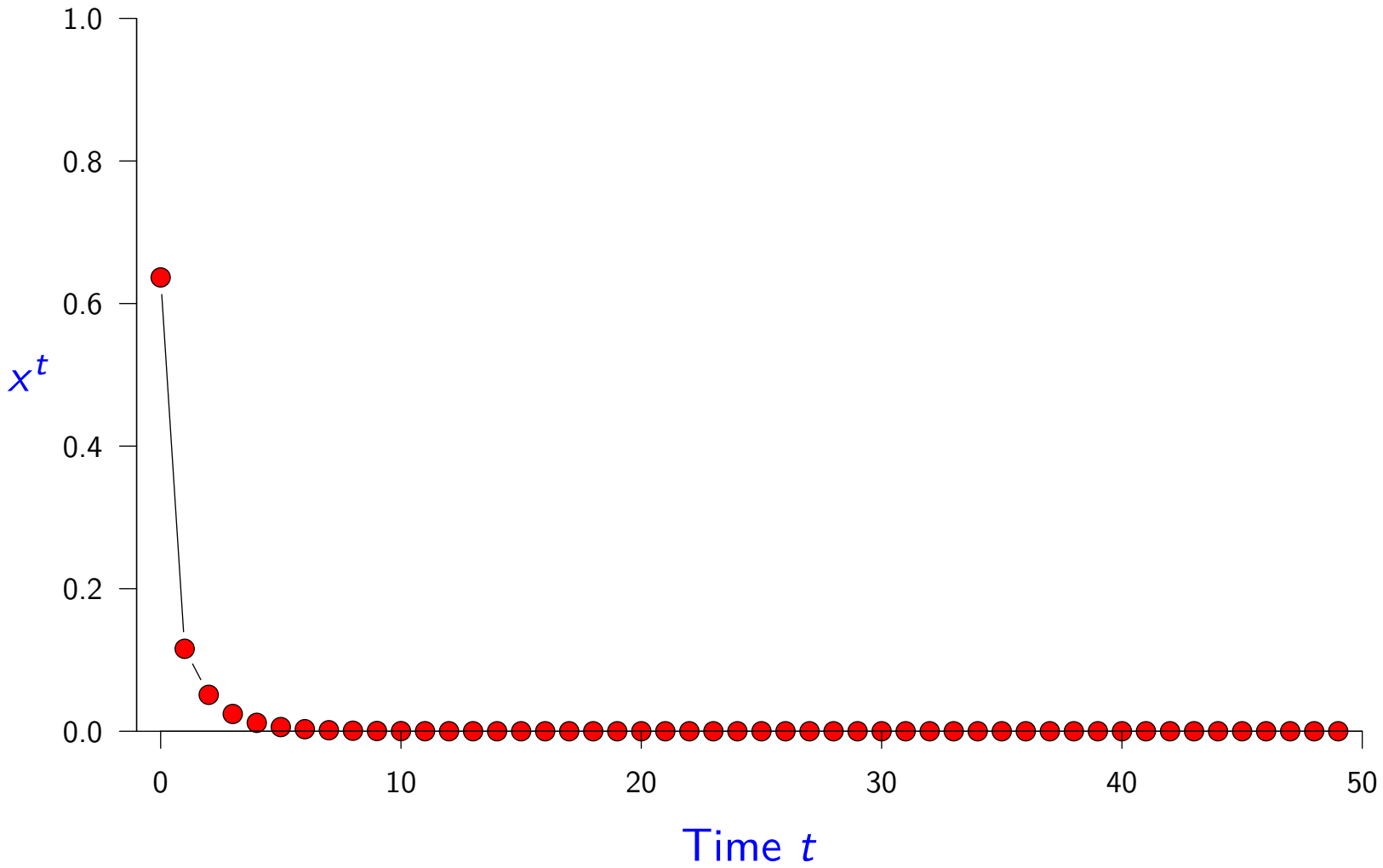
The Logistic Map

Logistic Map

- Simplest non-trivial *discrete time* population model for a single species (with *non-overlapping generations*) in a *single habitat patch*.
- Time: $t = 0, 1, 2, 3, \ldots$
- State: $x \geq 0$ (population density)
- Population density at time t is x^t. Solutions are sequences:
$$x^0,\ x^1,\ x^2,\ \ldots$$
- $x^{t+1} = F(x^t)$ for some *reproduction function* $F(x)$.
- For ***logistic map***: $F(x) = rx(1-x)$, so $x^{t+1} = rx^t(1-x^t)$.
$x^{t+1} = [r(1-x^t)]x^t \implies r$ is *maximum fecundity* (which is achieved in limit of very small population density).
Also, for $0 \leq r \leq 4$:
$x^0 \in [0,1] \implies F(x^0) \in [0,1] \implies x^t \in [0,1]\ \forall t \in \mathbb{N}$.
- What kinds of dynamics are possible for the Logistic Map?

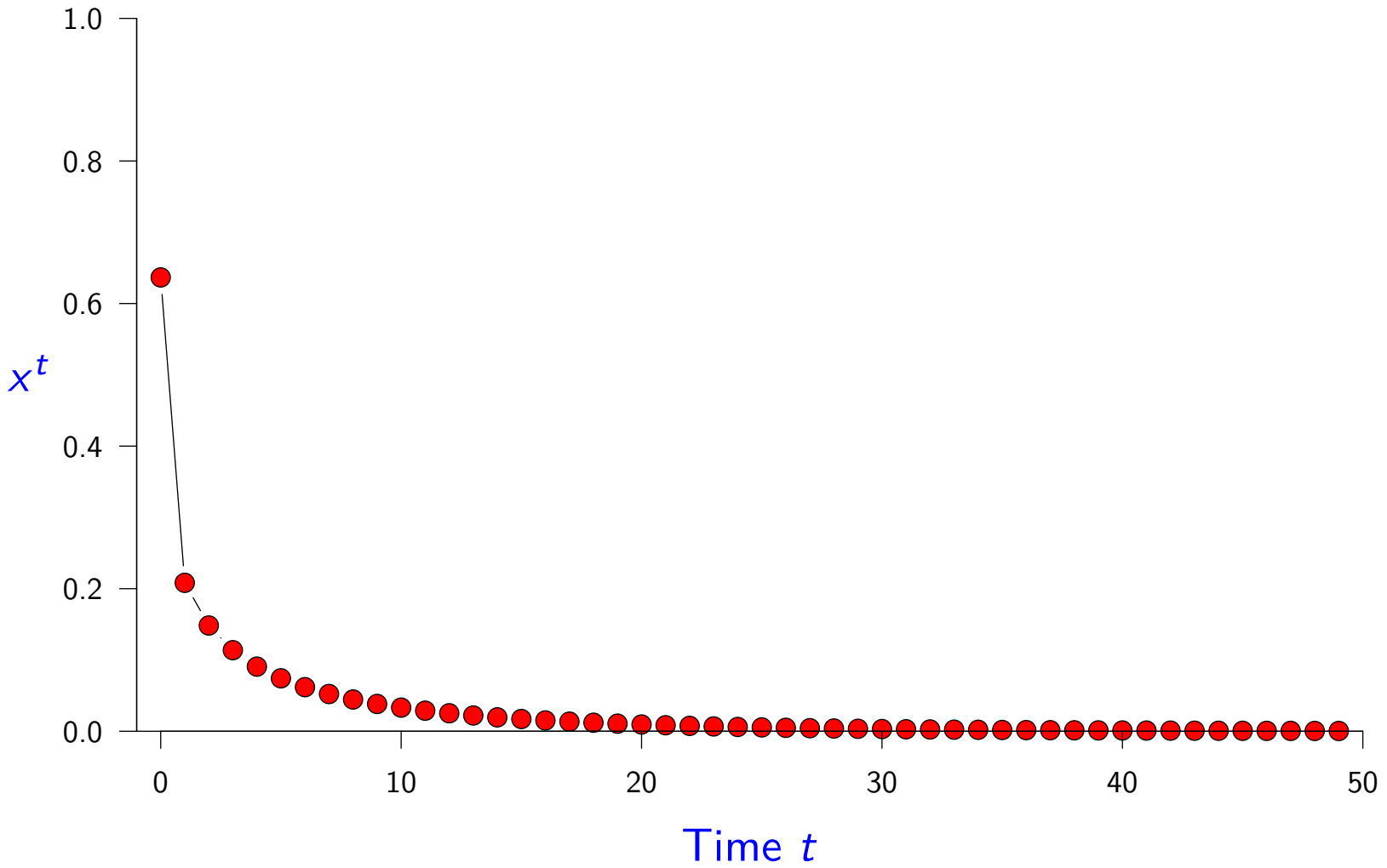
Logistic Map Time Series, $r = 0.5$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 0.5, \quad x_0 = 0.63662$$



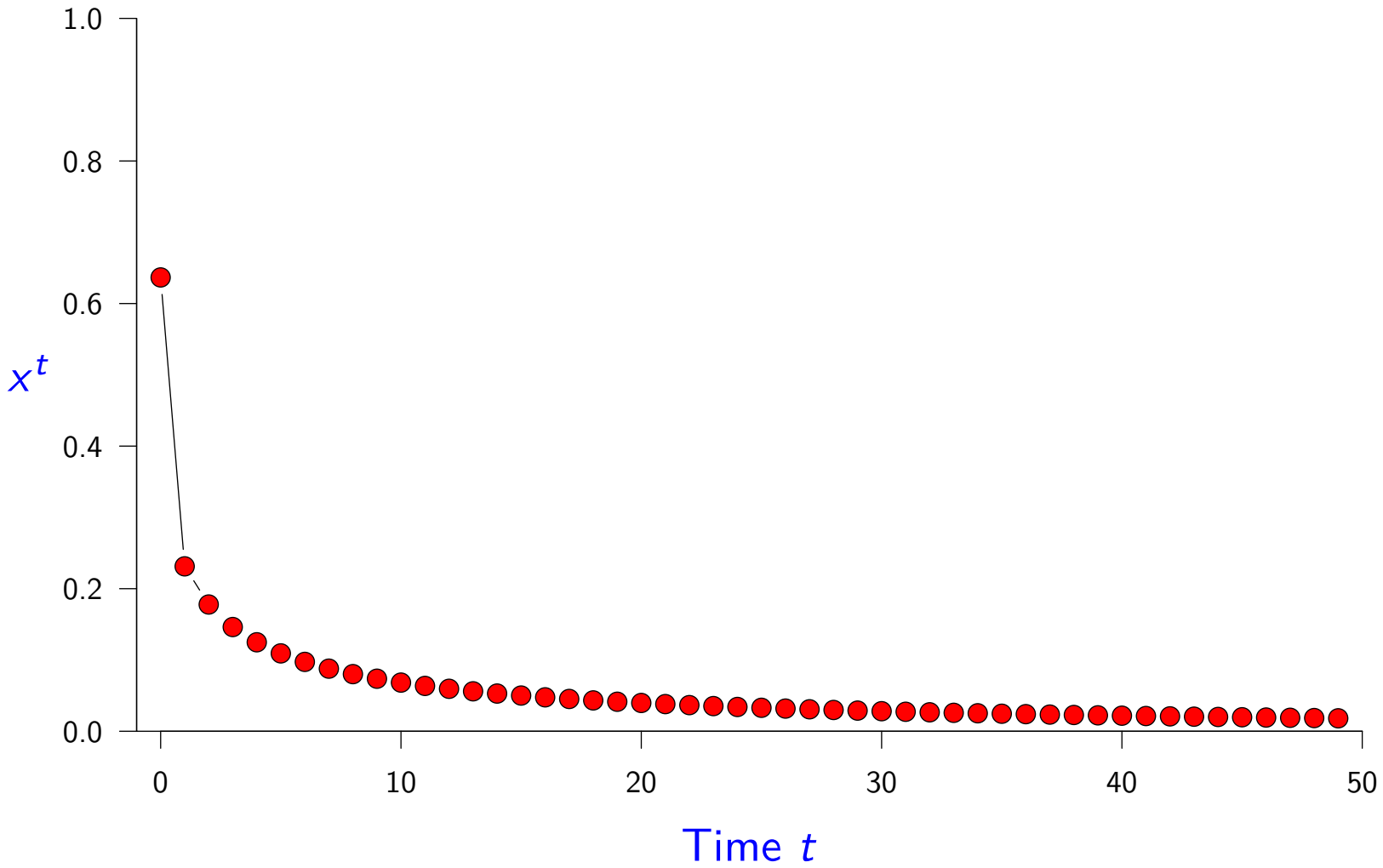
Logistic Map Time Series, $r = 0.9$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 0.9, \quad x_0 = 0.63662$$



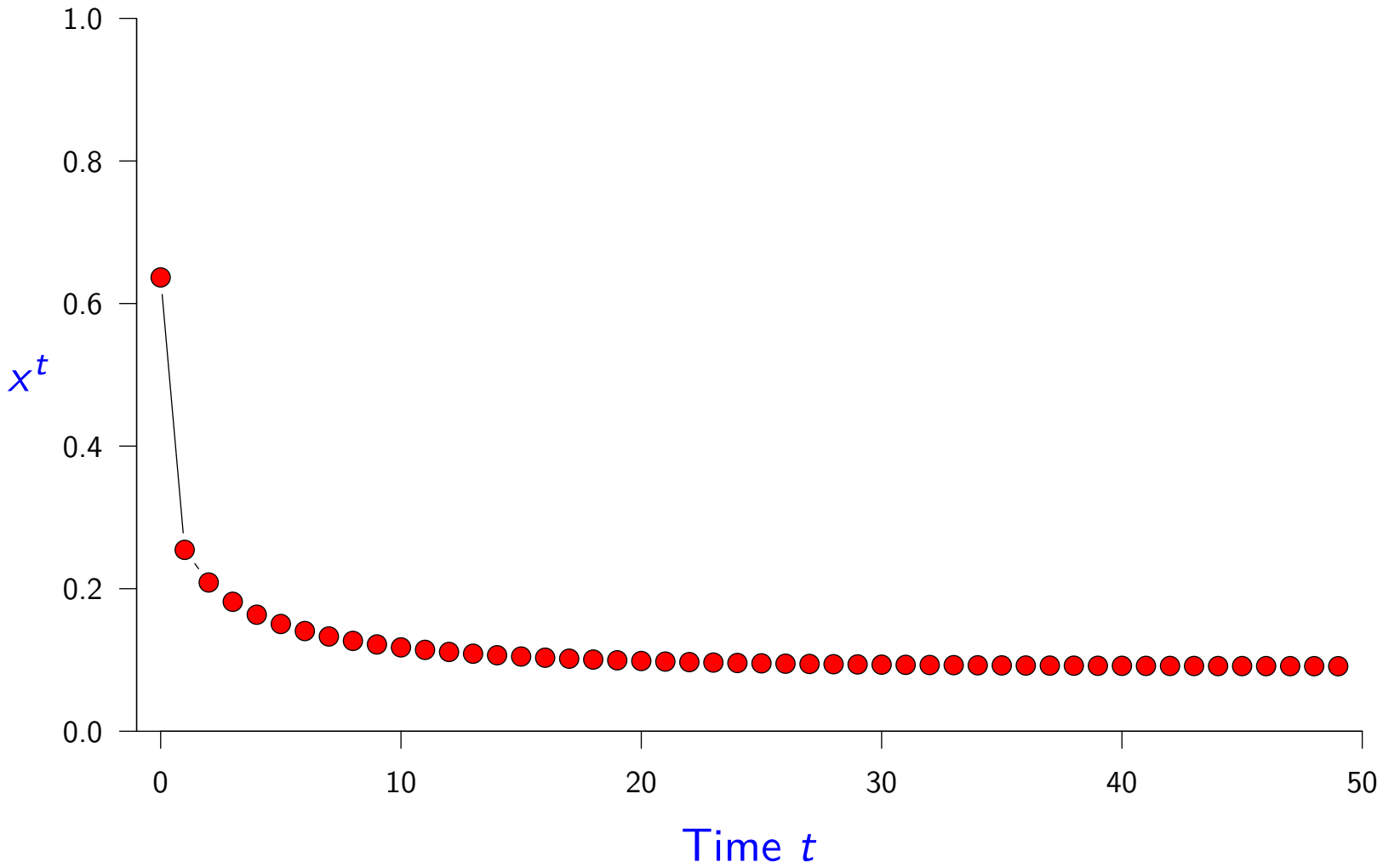
Logistic Map Time Series, $r = 1$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 1, \quad x_0 = 0.63662$$



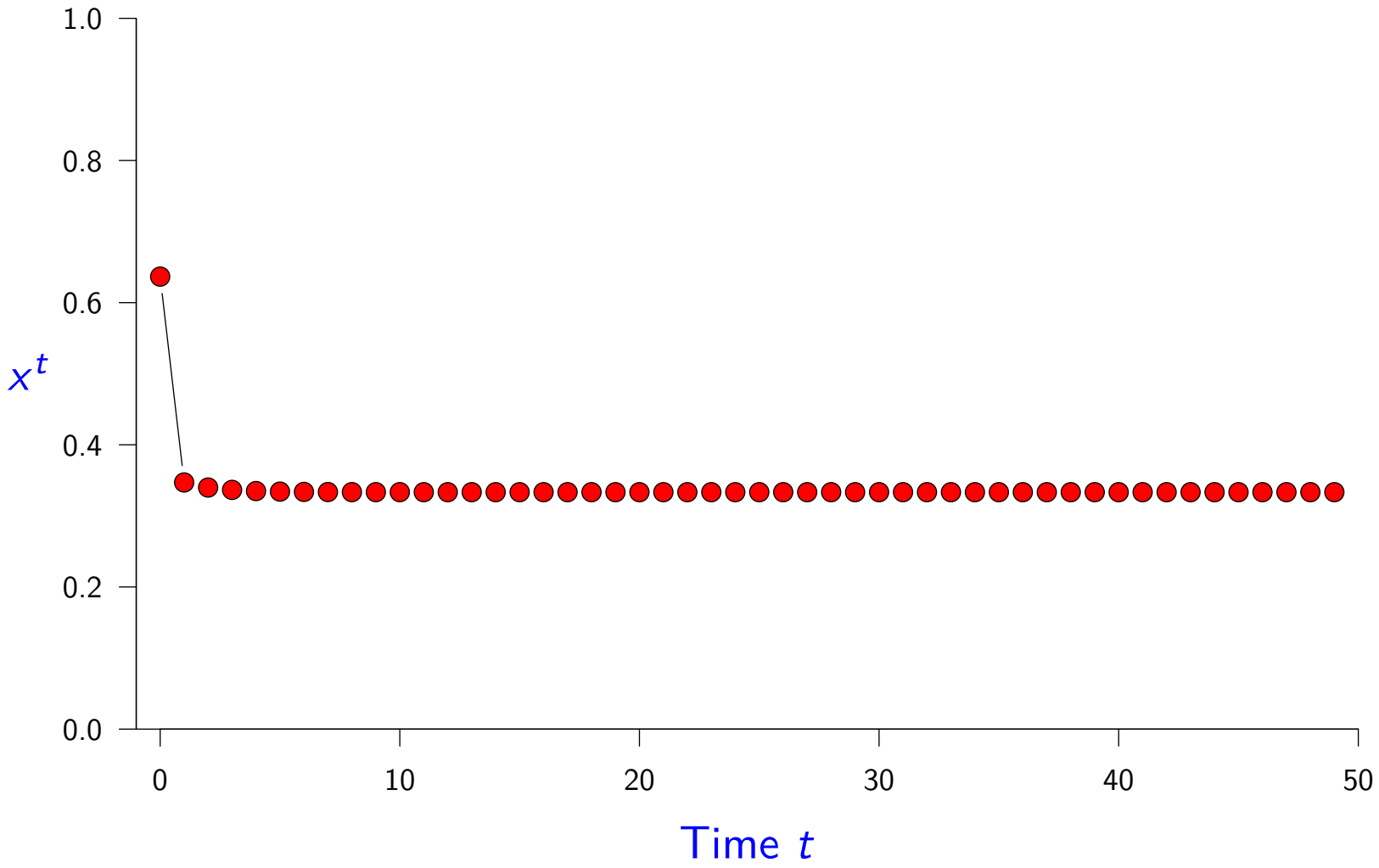
Logistic Map Time Series, $r = 1.1$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 1.1, \quad x_0 = 0.63662$$



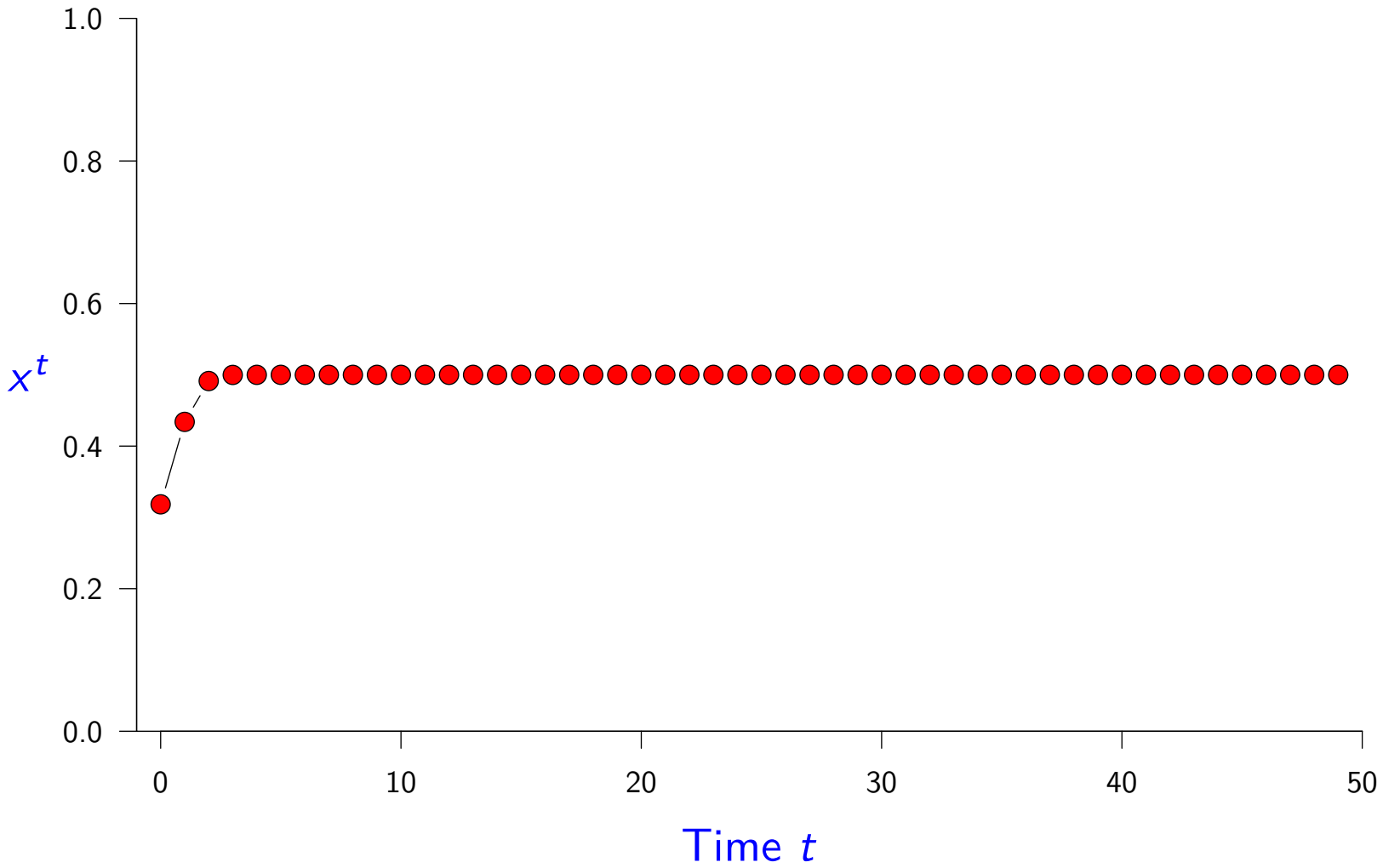
Logistic Map Time Series, $r = 1.5$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 1.5, \quad x_0 = 0.63662$$



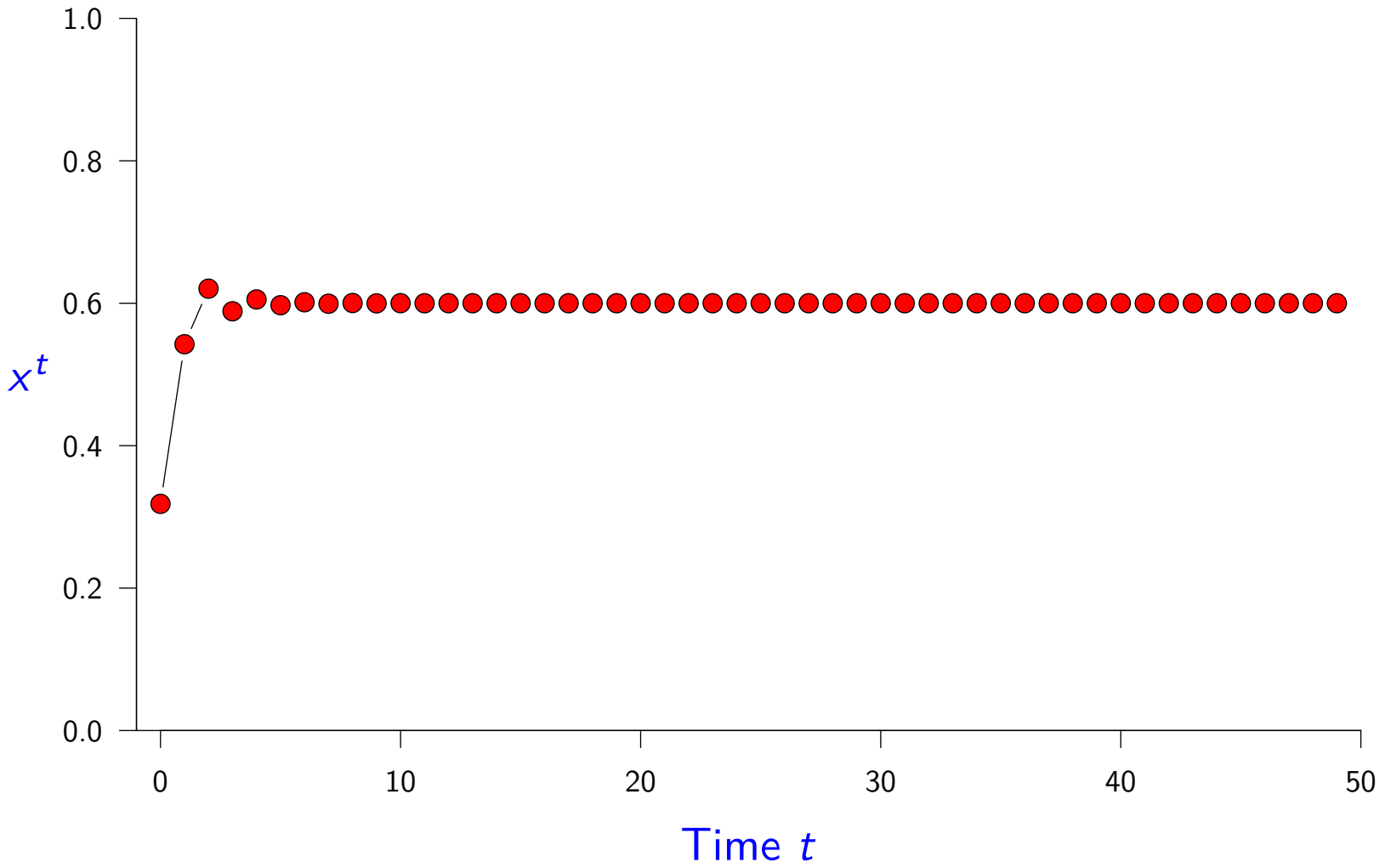
Logistic Map Time Series, $r = 2$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 2, \quad x_0 = 0.31831$$



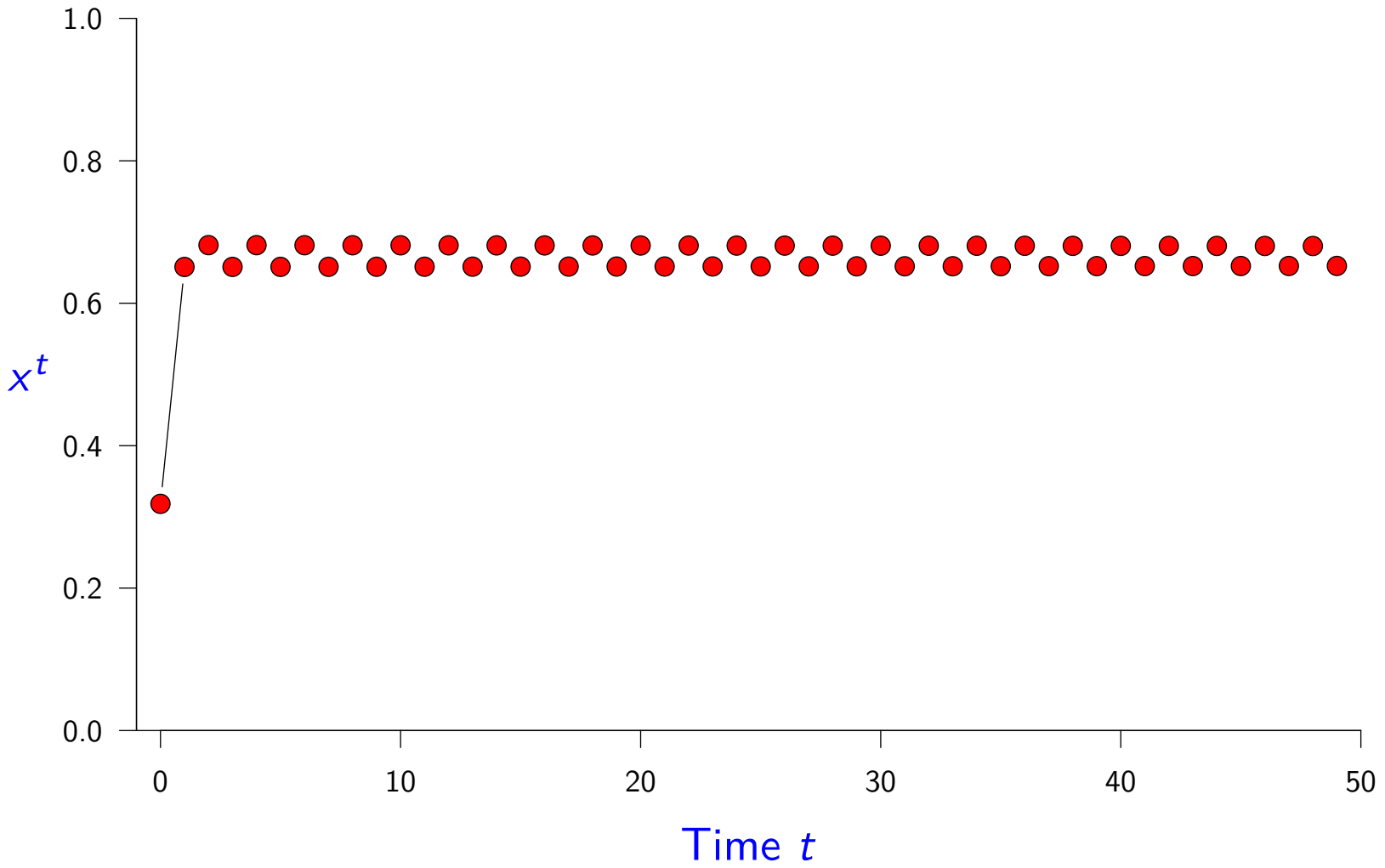
Logistic Map Time Series, $r = 2.5$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 2.5, \quad x_0 = 0.31831$$



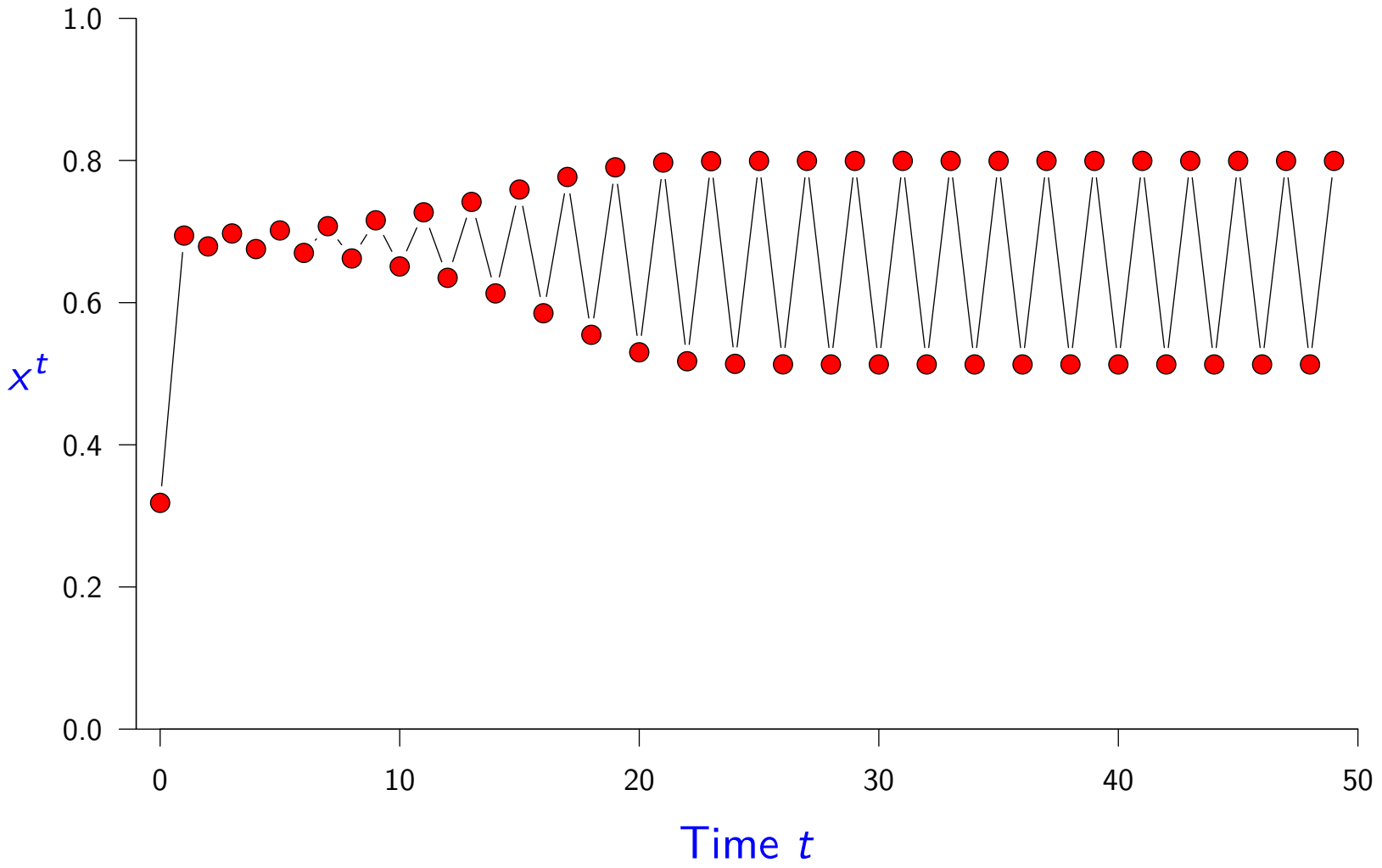
Logistic Map Time Series, $r = 3$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 3, \quad x_0 = 0.31831$$



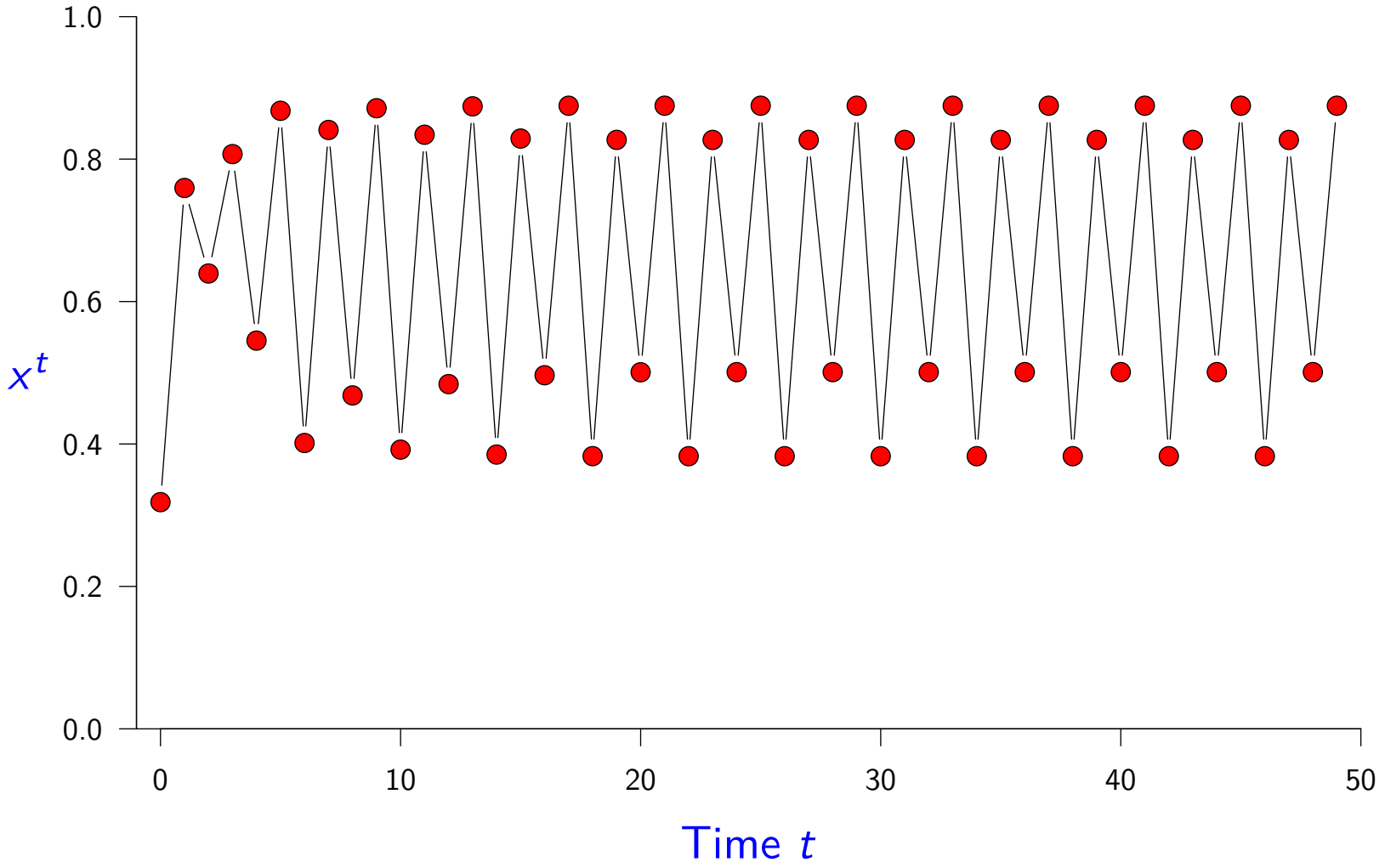
Logistic Map Time Series, $r = 3.2$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 3.2, \quad x_0 = 0.31831$$



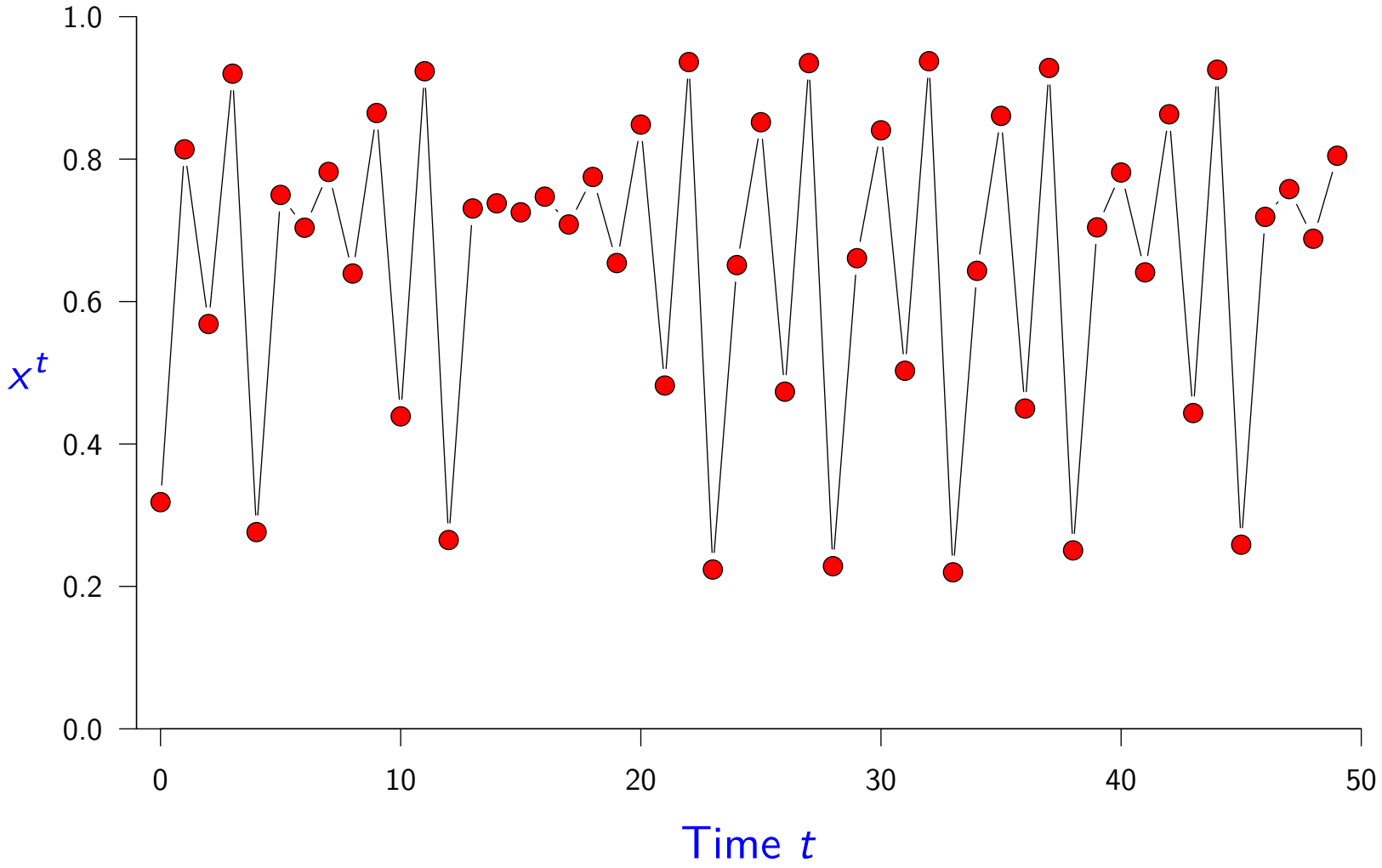
Logistic Map Time Series, $r = 3.5$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 3.5, \quad x_0 = 0.31831$$



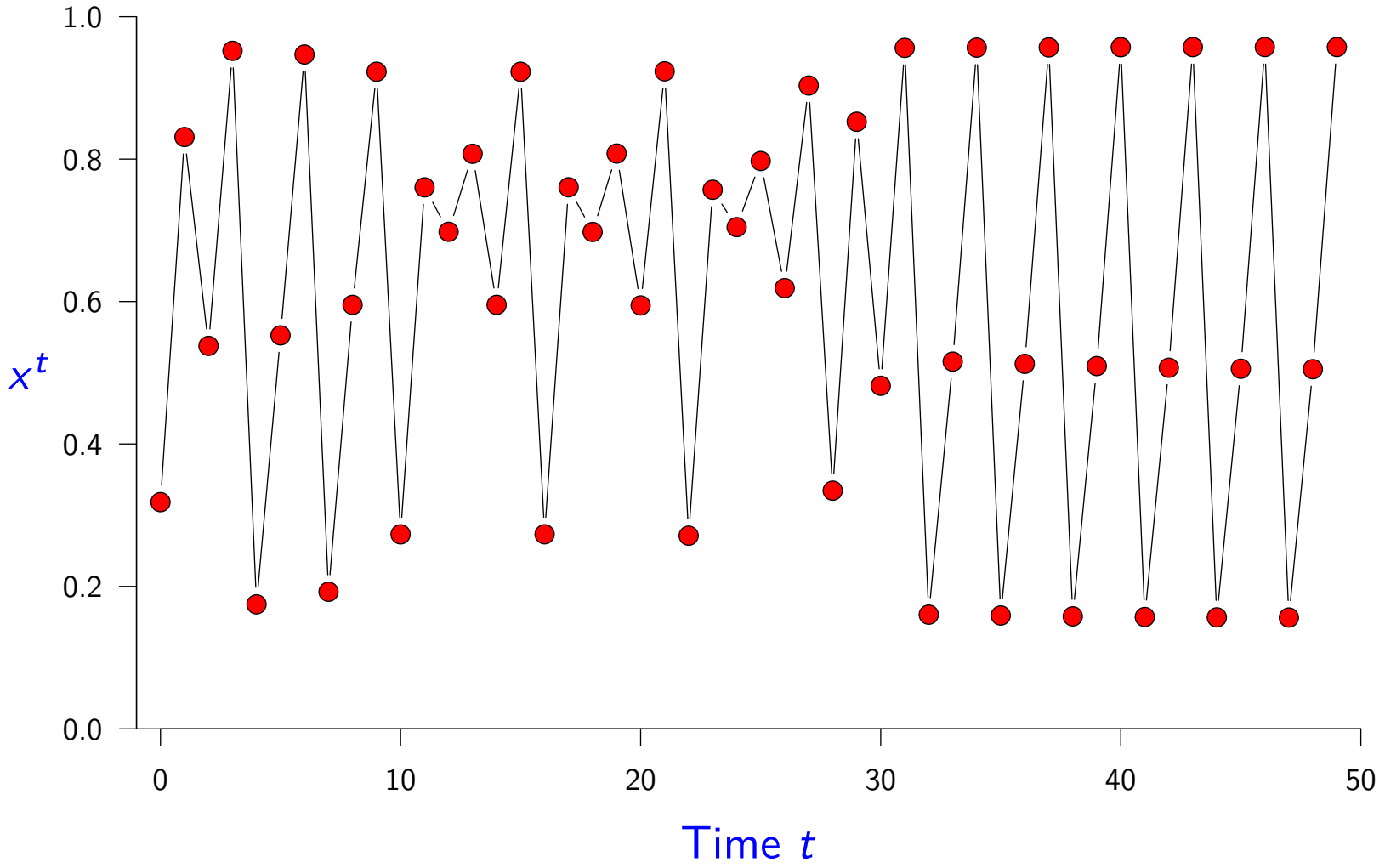
Logistic Map Time Series, $r = 3.75$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 3.75, \quad x_0 = 0.31831$$



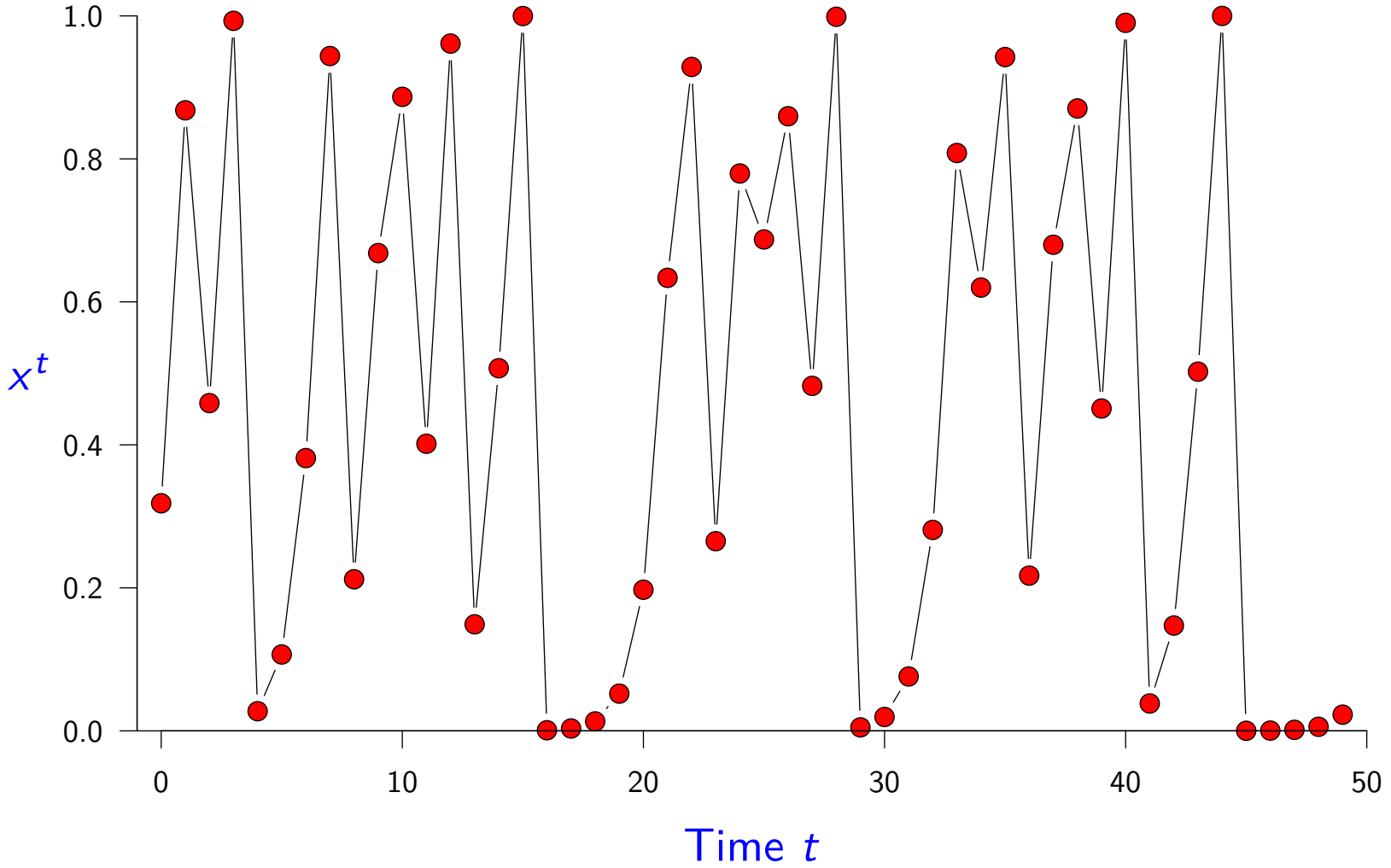
Logistic Map Time Series, $r = 3.83$

$$x^{t+1} = rx^t(1 - x^t), \quad r = 3.83, \quad x_0 = 0.31831$$



Logistic Map Time Series, $r = 4$

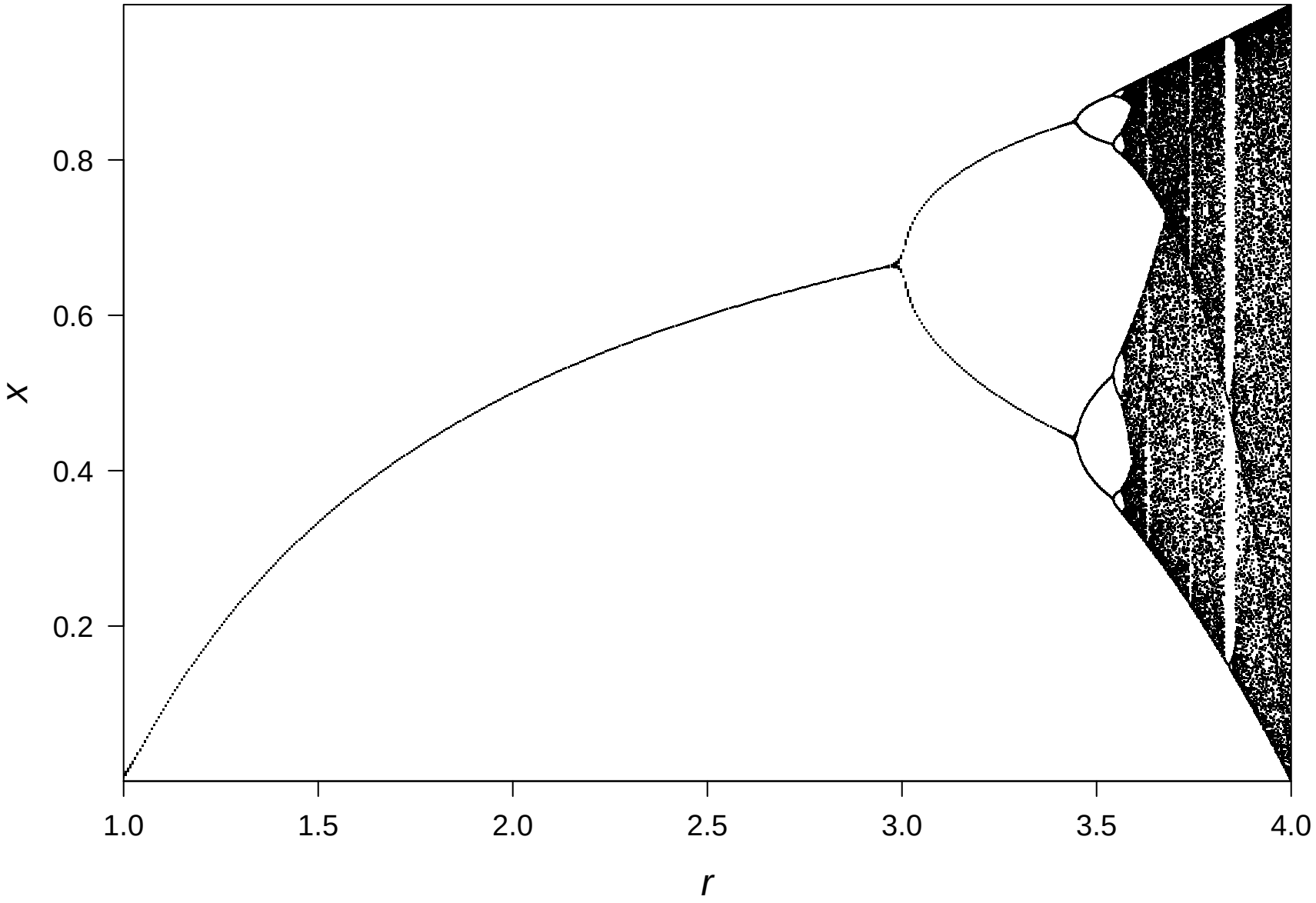
$$x^{t+1} = rx^t(1 - x^t), \quad r = 4, \quad x_0 = 0.31831$$



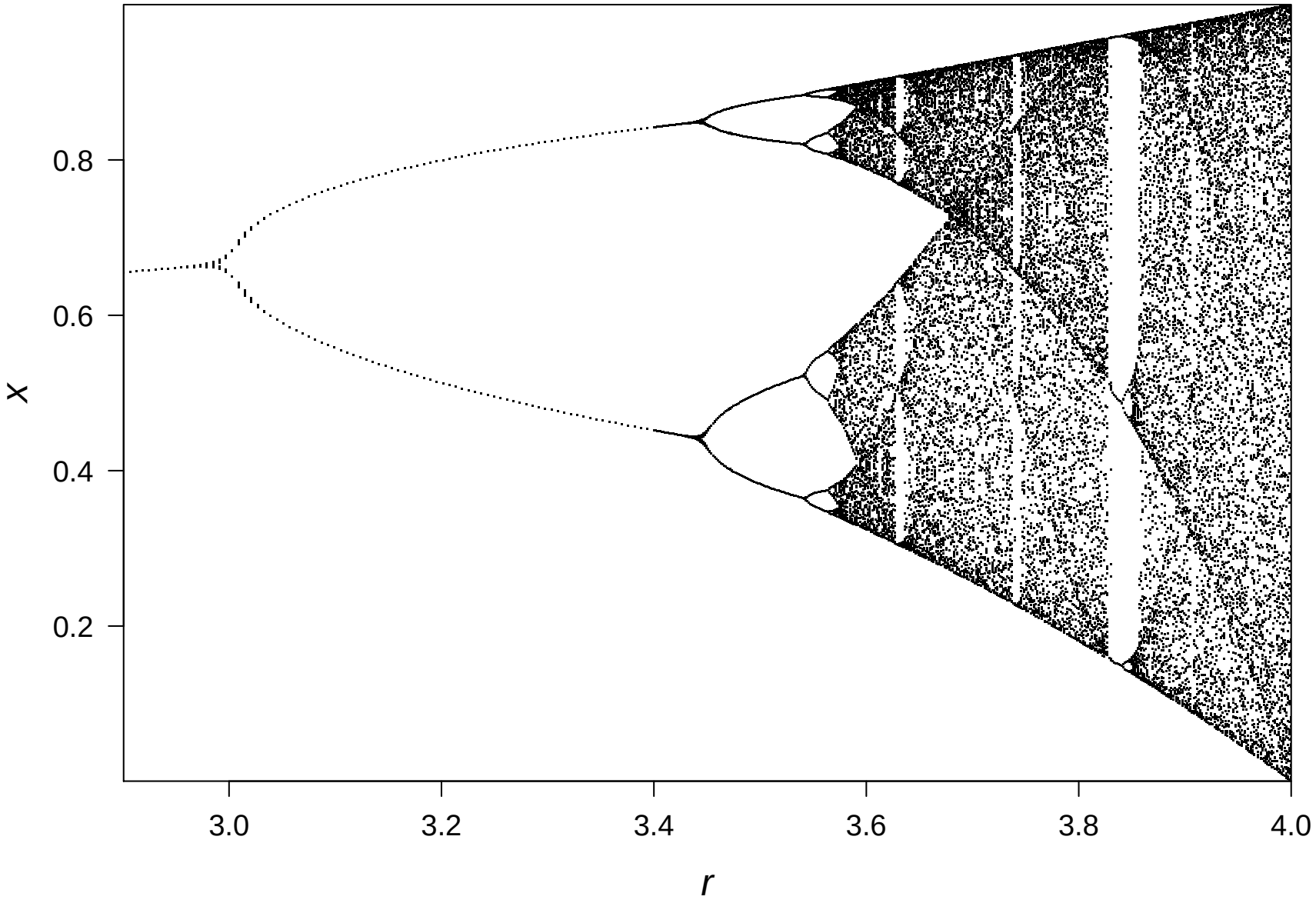
Logistic Map Summary

- Time series show:
  - $r \leq 1 \implies$ Extinction.
  - $1 < r < 3 \implies$ Persistence at equilibrium.
  - $r > 3 \implies$ period doubling cascade to chaos, then appearance of cycles of all possible lengths, and more chaos, ...
- How can we summarize this in a diagram?
  - Bifurcation diagram (wrt r).
  - Ignore transient behaviour: just show attractor.

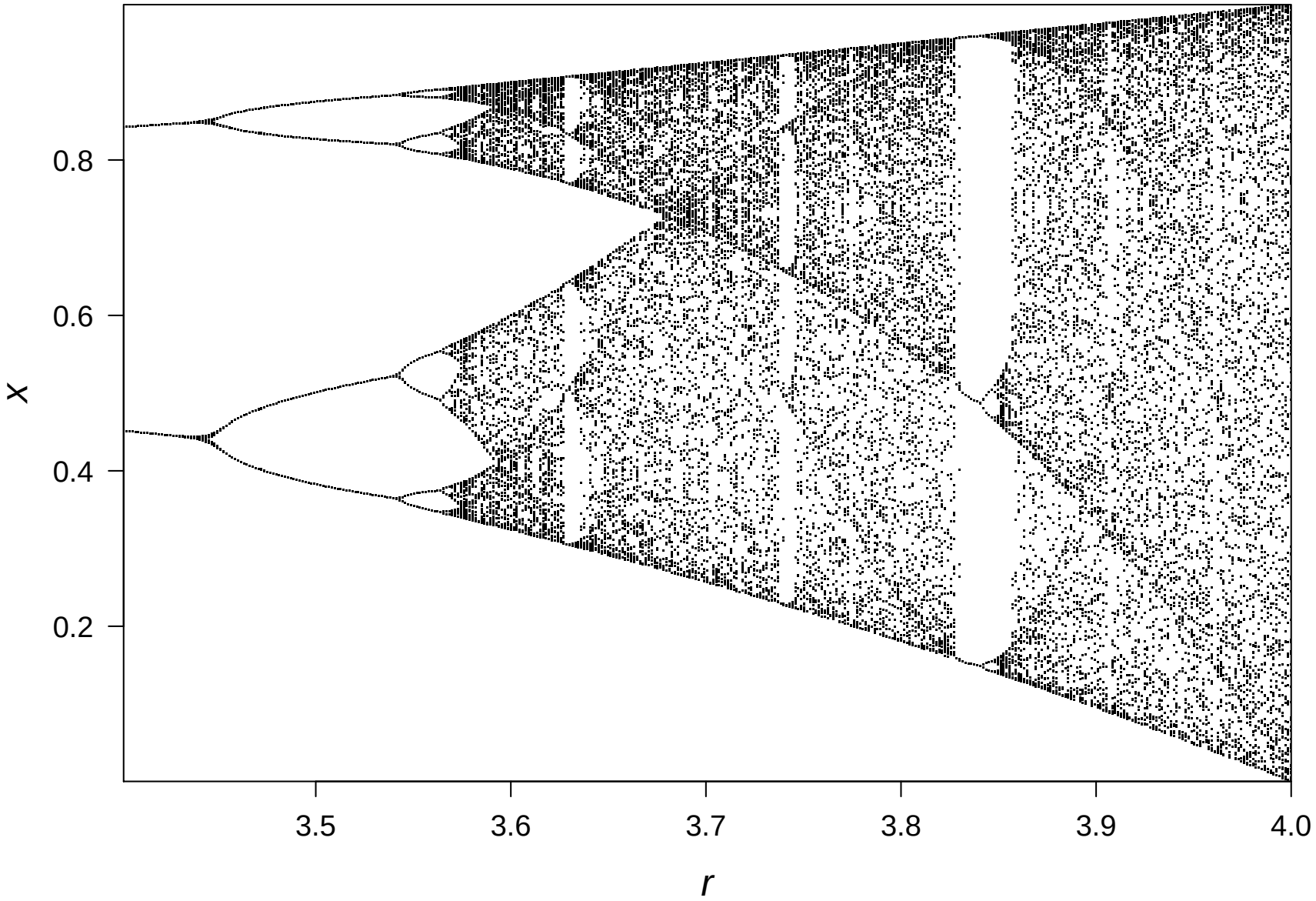
Logistic Map, $F(x) = rx(1-x), \quad 1 \leq r \leq 4$



Logistic Map, $F(x) = rx(1-x), \quad 2.9 \leq r \leq 4$



Logistic Map, $F(x) = rx(1-x), \quad 3.4 \leq r \leq 4$



Logistic Map as a Tool to Investigate Synchrony

- Very simple single-patch model: only one state variable.
- Displays all kinds of dynamics from GAS equilibrium, to periodic orbits, to chaos.
  - This was *extremely surprising* to population biologists and mathematicians in the 1970s.

  May RM (1976) "Simple mathematical models with very complicated dynamics" *Nature* **261**, 459–467
- Easier to work with logistic map as single patch dynamics than SIR or SEIR model.
- Can still understand how synchrony works conceptually.
- Now we are ready for the ...

...*Mathematics of Synchrony* ...

Mathematics of Synchrony

- System comprised of isolated *patches*
  e.g., cities, labelled $i = 1, \ldots, n$
- *State* of system in patch i specified by $\mathbf{x}_i$
  e.g., $\mathbf{x}_i = (S_i, E_i, I_i, R_i)$
- Connectivity of patches specified by a *dispersal matrix*
  $\mathrm{M} = (m_{ij})$
- System is *coherent* (perfectly synchronous) if the state is the same in all patches
  i.e., $\mathbf{x}_1 = \mathbf{x}_2 = \cdots = \mathbf{x}_n$

Illustrative example: logistic metapopulation

- *Single patch model:* $x^{t+1} = F(x^t)$
- *Reproduction function:* $F(x) = rx(1-x)$
- *Multi-patch model:* $x_i^{t+1} = \sum_{j=1}^{n} m_{ij} F(x_j{}^t)$

$$\textit{i.e.,} \quad \begin{pmatrix} x_1{}^{t+1} \\ \vdots \\ x_n{}^{t+1} \end{pmatrix} = \begin{pmatrix} m_{11} & \cdots & m_{1n} \\ \vdots & \ddots & \vdots \\ m_{n1} & \cdots & m_{nn} \end{pmatrix} \begin{pmatrix} F(x_1{}^t) \\ \vdots \\ F(x_n{}^t) \end{pmatrix}$$

where $\mathrm{M} = (m_{ij})$ is *dispersal matrix*.

- *Colour coding of matrix indices:*
  - row indices are red
  - column indices are cyan

Basic properties of dispersal matrices $M = (m_{ij})$

Discrete-time *metapopulation* model:

$$x_i^{t+1} = \sum_{j=1}^{n} m_{ij} F(x_j^t), \qquad i = 1, 2, \ldots, n.$$

- m_{ij} = *proportion* of population in patch j that disperses to patch i.
- $\therefore\ 0 \le m_{ij} \le 1$ for all i and j
  (*each m_{ij} is non-negative and at most 1*)
- Total proportion that leaves or stays in patch j: $\sum_{i=1}^{n} m_{ij}$
  (*sum of column j*)
- $\therefore\ \sum_{i=1}^{n} m_{ij} \le 1$ (*every column sums to at most 1*)

Could be < 1 if some individuals are lost (die) while dispersing.

Basic properties of dispersal matrices $\mathrm{M} = (m_{ij})$

Discrete-time *metapopulation* model:

$$x_i^{t+1} = \sum_{j=1}^{n} m_{ij} F(x_j^t), \qquad i = 1, 2, \ldots, n.$$

Definition (No loss dispersal matrix)

An $n \times n$ matrix $\mathrm{M} = (m_{ij})$ is said to be a ***no loss dispersal matrix*** if all its entries are non-negative ($m_{ij} \geq 0$ for all i and j) and its column sums are all 1, *i.e.*,

$$\sum_{i=1}^{n} m_{ij} = 1\,, \qquad \text{for each } j = 1, \ldots, n.$$

- The dispersal process is "conservative" in this case.
- A no loss dispersal matrix is also said to be "column stochastic".

Notation for coherent states

Discrete-time *metapopulation* model:

$$x_i^{t+1} = \sum_{j=1}^{n} m_{ij} F(x_j^t), \qquad i = 1, 2, \ldots, n.$$

- State at time t is $\mathbf{x}^t = (x_1^t, \ldots, x_n^t) \in \mathbb{R}^n$.
- If state $\mathbf{x}$ is *coherent*, then for some $x \in \mathbb{R}$ we have

$$\begin{aligned} \mathbf{x} &= (x_1, x_2, \ldots, x_n) \\ &= (x, x, \ldots, x) = x(1, 1, \ldots, 1) \end{aligned}$$

- For convenience, define

$$\boldsymbol{e} = (1, 1, \ldots, 1) \in \mathbb{R}^n$$

so any coherent state can be written $x\boldsymbol{e}$*, for some* $x \in \mathbb{R}$*.*

Constraint on row sums of dispersal matrix M

Lemma (Row sums are the same)

If all initially coherent states remain coherent then the row sums of the dispersal matrix are all the same.

Proof.

Suppose initially coherent states remain coherent, *i.e.*,
$\mathbf{x}^t = a\mathbf{e} \implies \mathbf{x}^{t+1} = b\mathbf{e}$ for some $b \in \mathbb{R}$.
Choose a such that $F(a) \neq 0$. Then

$$x_i^{t+1} = b = \sum_{j=1}^{n} m_{ij} F(x_j^t) = \sum_{j=1}^{n} m_{ij} F(a) = F(a) \sum_{j=1}^{n} m_{ij}$$

$$\implies \sum_{j=1}^{n} m_{ij} = \frac{b}{F(a)} \qquad \text{(independent of } i\text{)}$$

□

Constraint on row sums of dispersal matrix M

Lemma (Row sums are all 1)

If every solution $\{x^t\}$ of the single patch map $F(x)$ yields a coherent solution $\{x^t e\}$ of the full map then the row sums of the dispersal matrix are all 1.

Proof.

Suppose $\mathbf{x}^t = ae \implies \mathbf{x}^{t+1} = F(a)e$ and $F(a) \neq 0$. Then

$$x_i^{t+1} = F(a) = \sum_{j=1}^{n} m_{ij} F(x_j^t) = \sum_{j=1}^{n} m_{ij} F(a) = F(a) \sum_{j=1}^{n} m_{ij}$$

$$\implies \sum_{j=1}^{n} m_{ij} = 1 \qquad \text{(independent of } i\text{)}$$

□

Project

Project

You should be thinking about your **Project**...

- Settle on project topic ASAP...
- Remember your group must give an oral presentation of your project as well (in the last class).
- Classes after the midterm are NOT optional. Your group is expected to meet in class and take advantage of the instructor's presence to solve issues with your project.
- Project Notebook template is posted on project page.
- Feedback on project draft...
- Movie night?

Back to Space and Synchrony

Let's review what we've done so far on spatial models...

- Logistic metapopulation model
- Notion of coherence
- No-loss dispersal matrix M: column sums are all 1
- To retain homogeneous solutions: row sums are all 1

Simple examples of no loss dispersal matrices

- *Equal coupling:* a proportion m from each patch disperses uniformly among the other $n-1$ patches:

$$m_{ij} = \begin{cases} 1-m & i=j \\ m/(n-1) & i \neq j \end{cases}$$

- *Nearest-neighbour coupling on a ring:* a proportion m go to the two nearest patches:

$$m_{ij} = \begin{cases} 1-m & i=j \\ m/2 & i=j-1 \text{ or } j+1 \pmod{n} \\ 0 & \text{otherwise} \end{cases}$$

- Real dispersal patterns generally between these two extremes

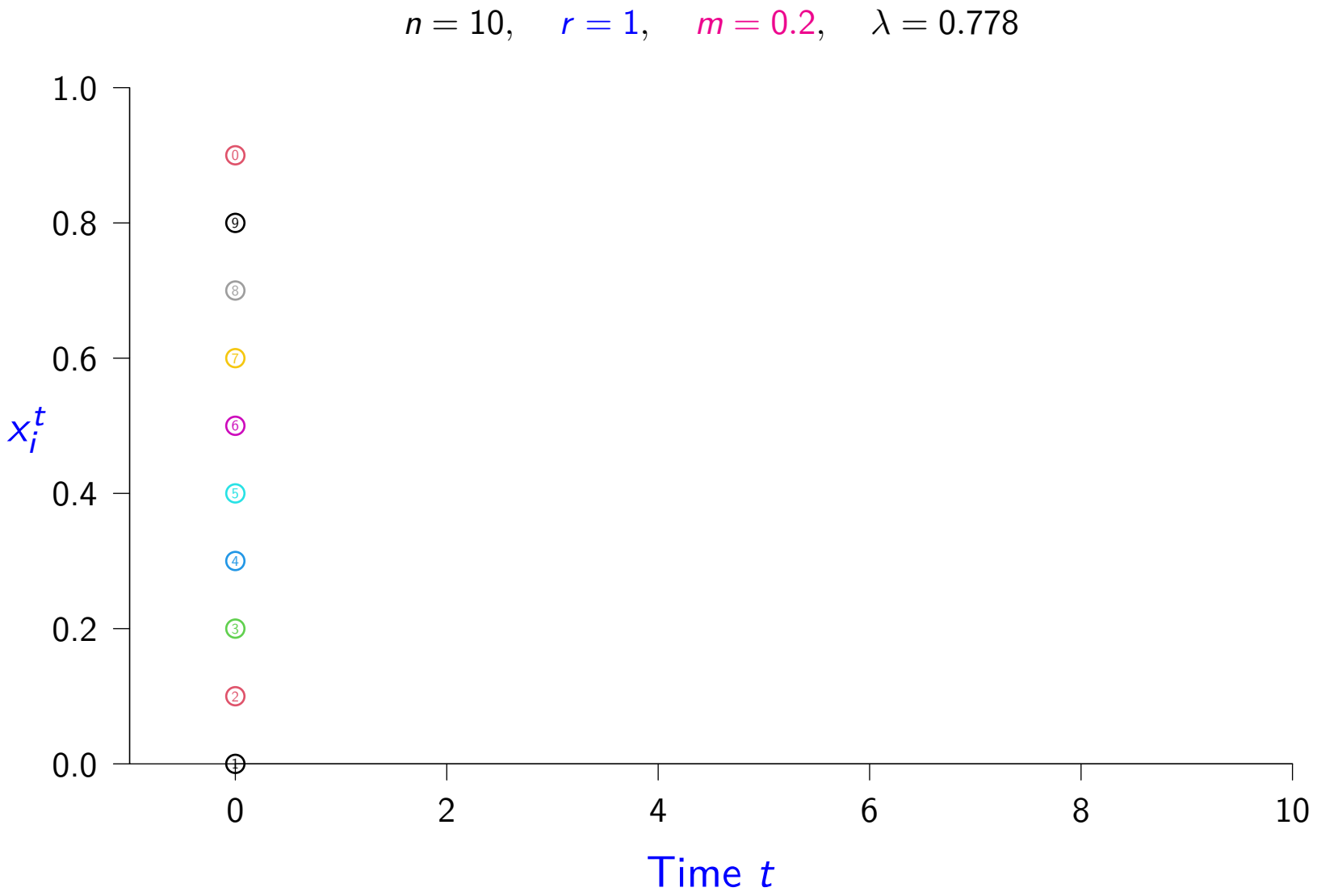
Key Question

- Can we find conditions on the dispersal matrix M, and/or the single patch reproduction function F, that guarantee (or preclude) coherence asymptotically (as $t \to \infty$)?
  - If so, then this sort of analysis should help to identify synchronizing vaccination strategies.

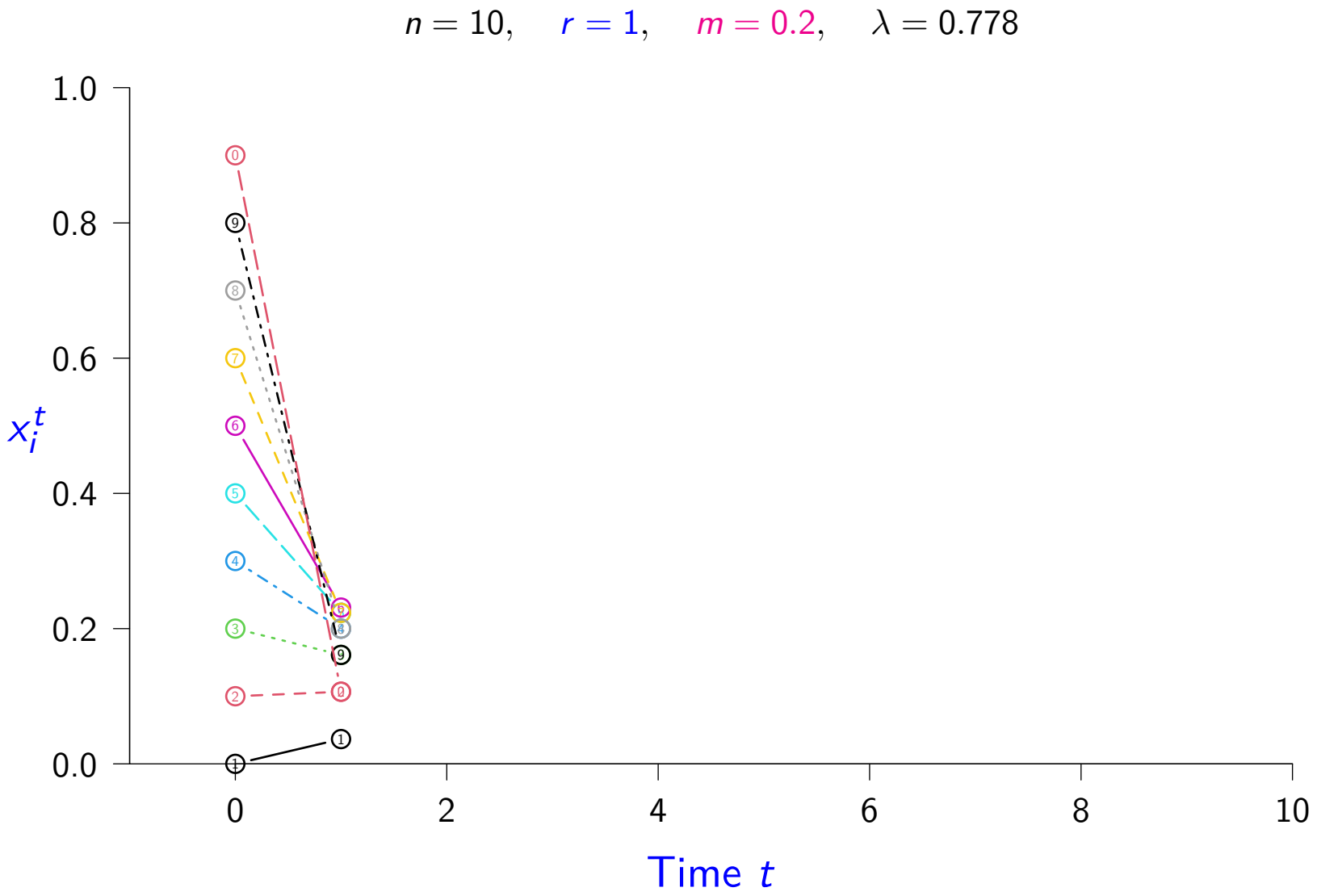
Exploratory simulations

- Let's try to build up some intuition by running simulations of a logistic metapopulation
  - Reproduction function $F(x) = r\,x\,(1-x)$
  - various levels of fecundity: $1 \leq r \leq 4$
  - $n = 10$ patches with equal coupling
  - various levels of connectivity: $0 \leq m \leq 1$

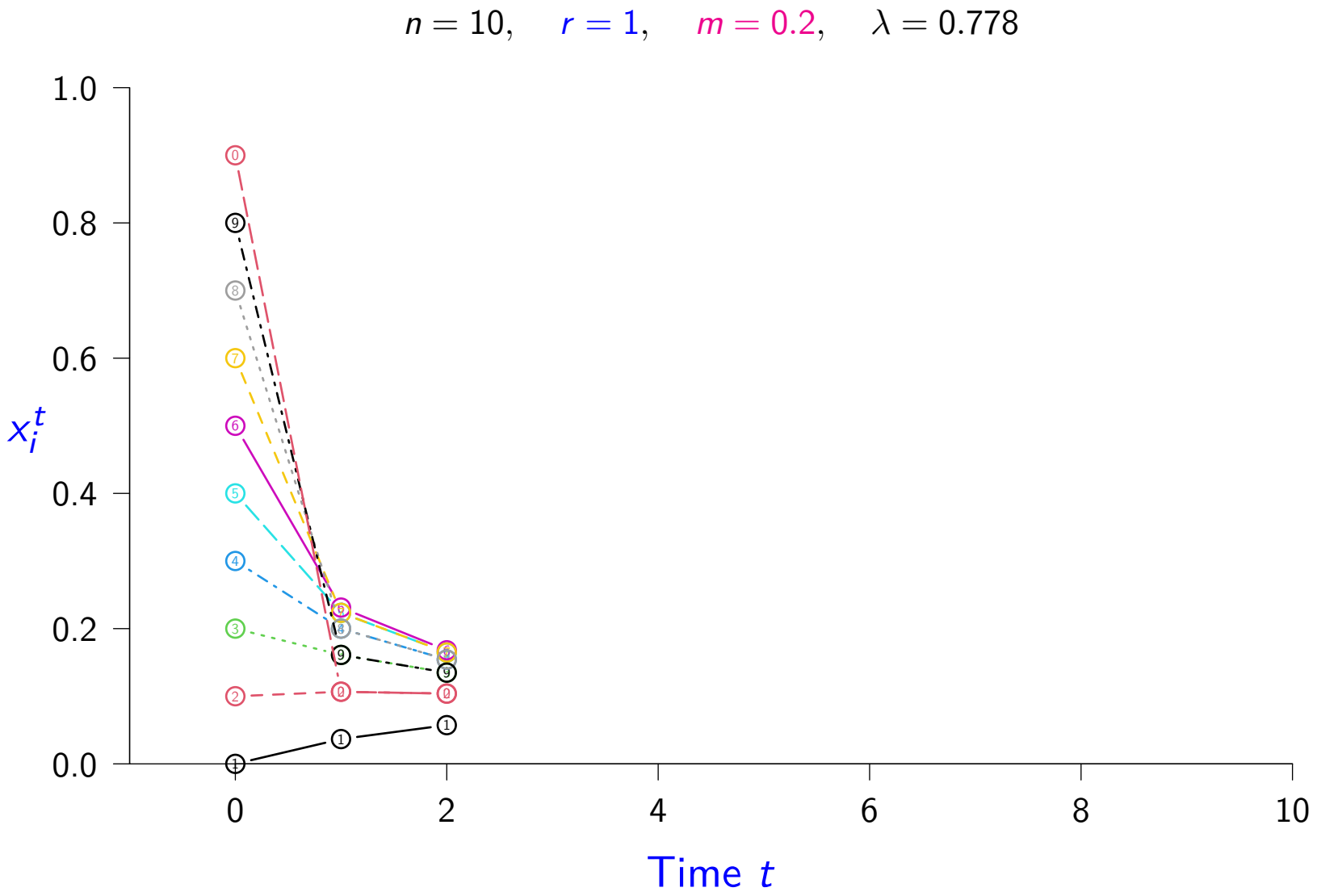
Logistic Metapopulation Simulation ($r = 1$, $m = 0.2$)



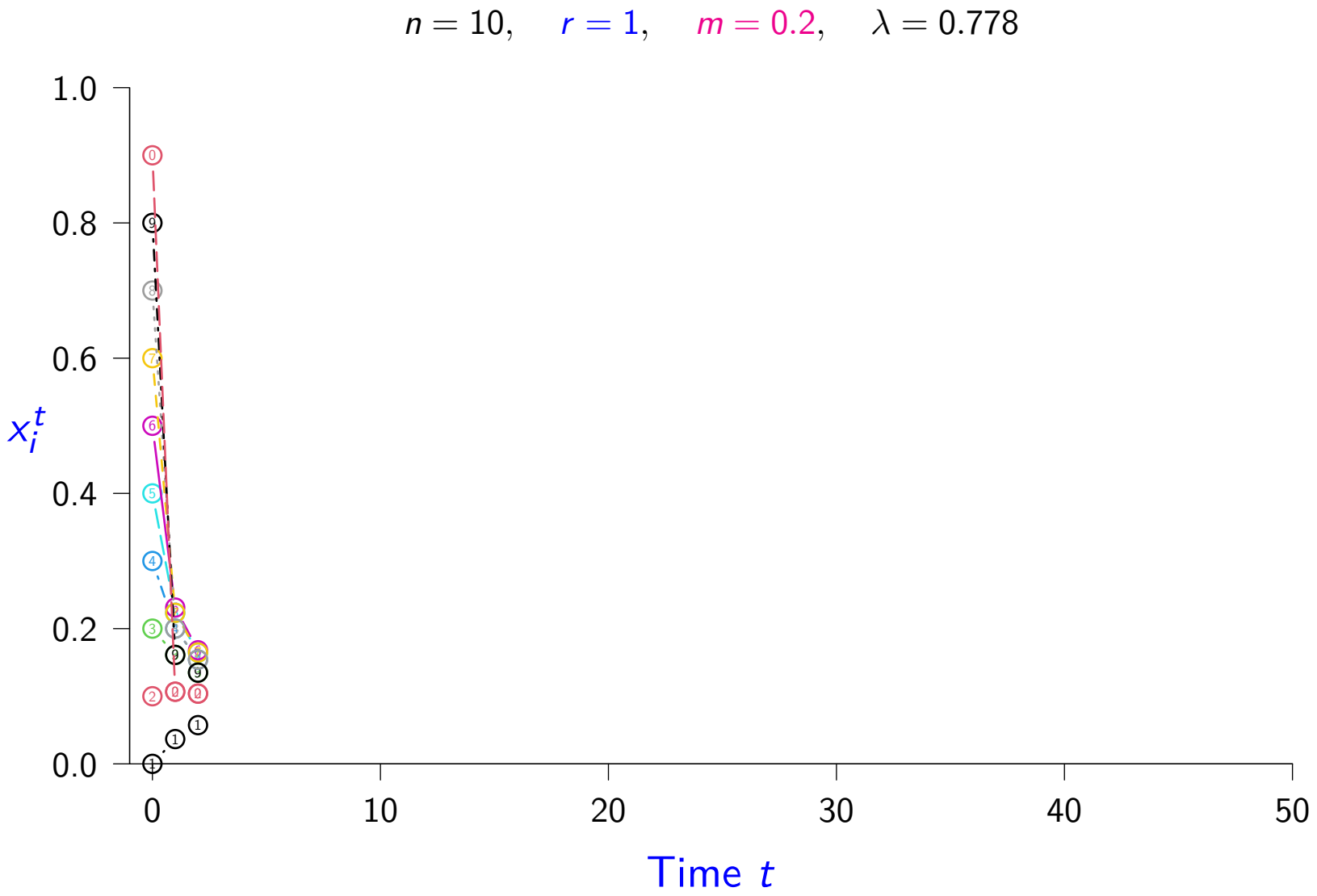
Logistic Metapopulation Simulation ($r = 1$, $m = 0.2$)



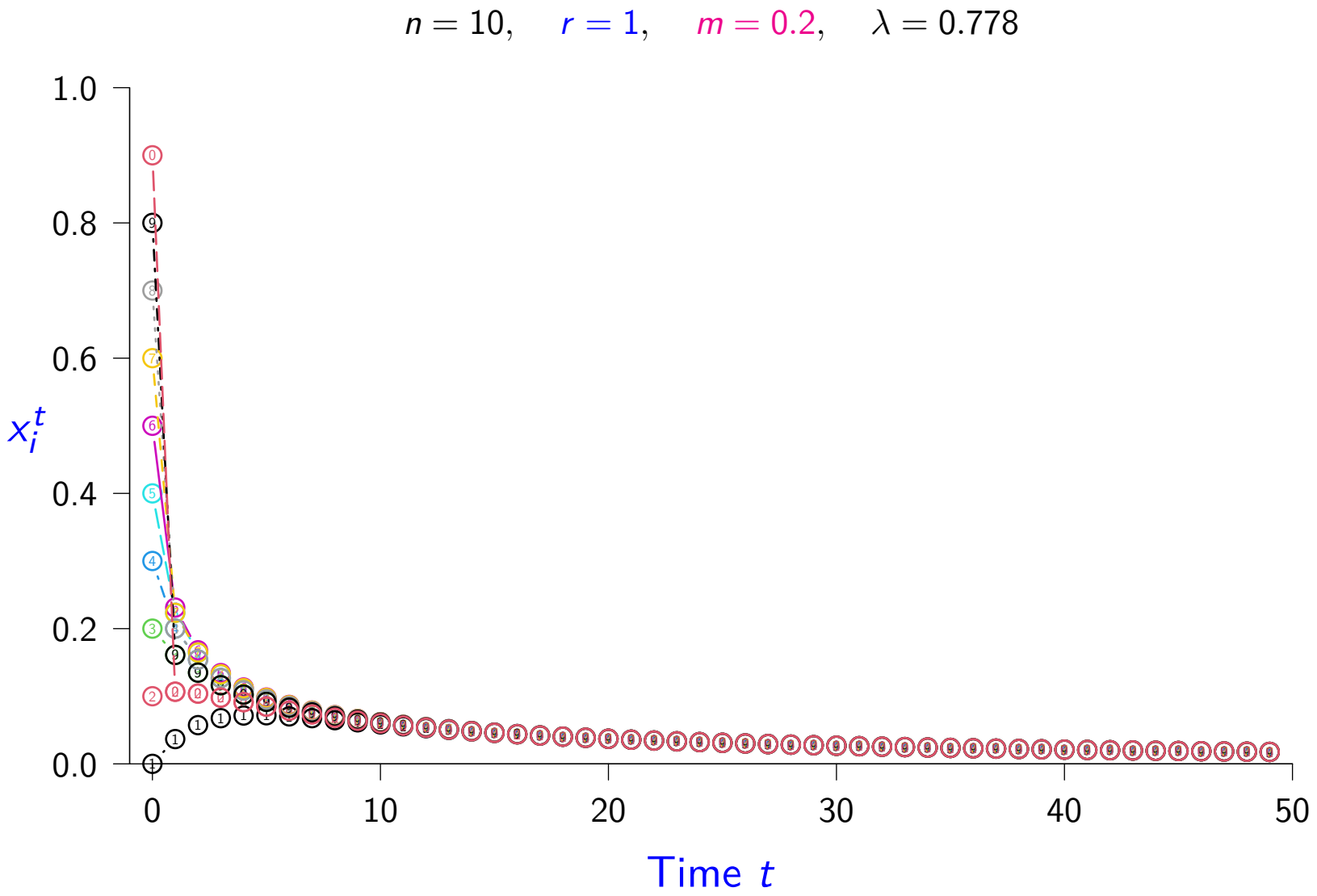
Logistic Metapopulation Simulation ($r = 1,\ m = 0.2$)



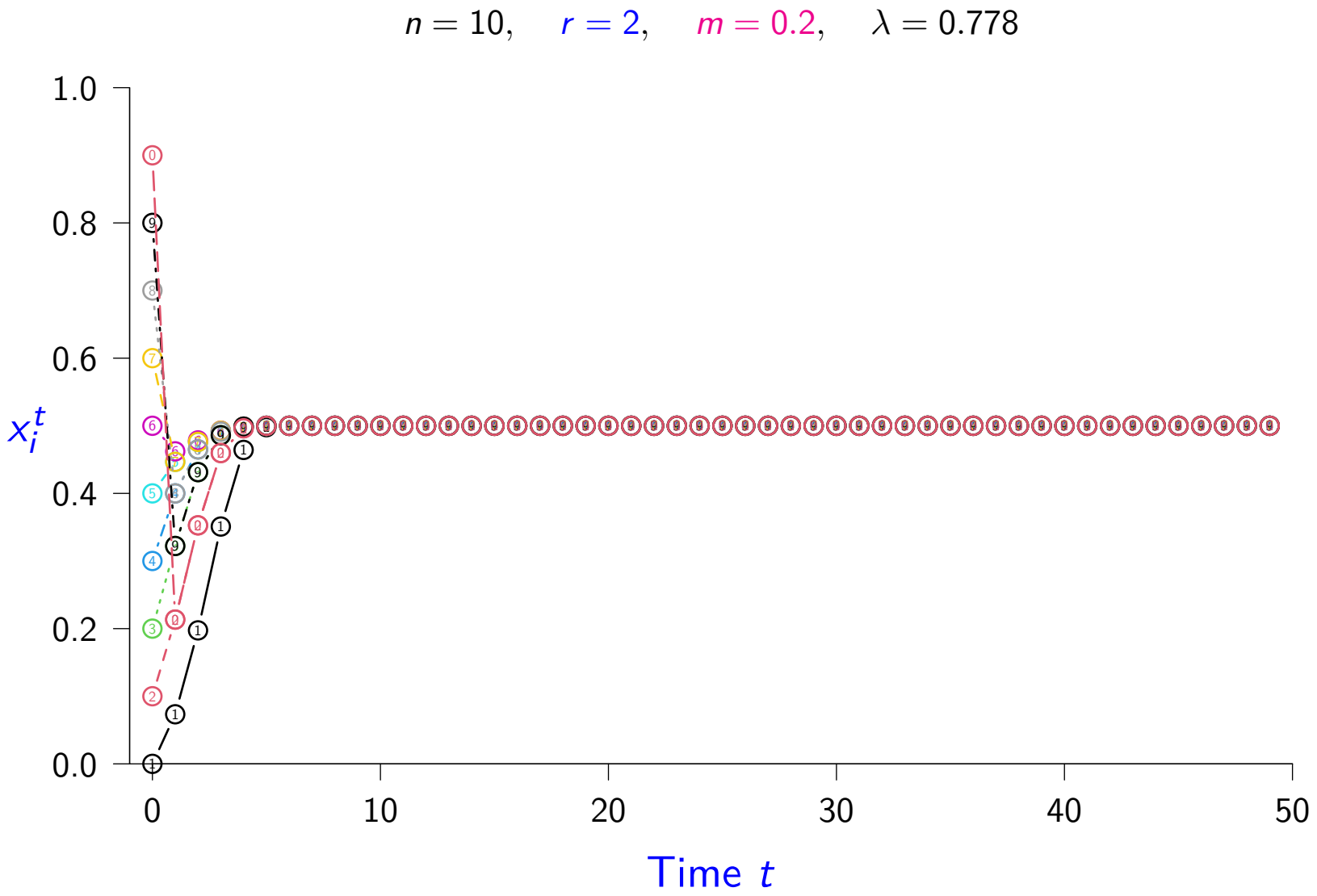
Logistic Metapopulation Simulation ($r = 1$, $m = 0.2$)



Logistic Metapopulation Simulation ($r = 1$, $m = 0.2$)

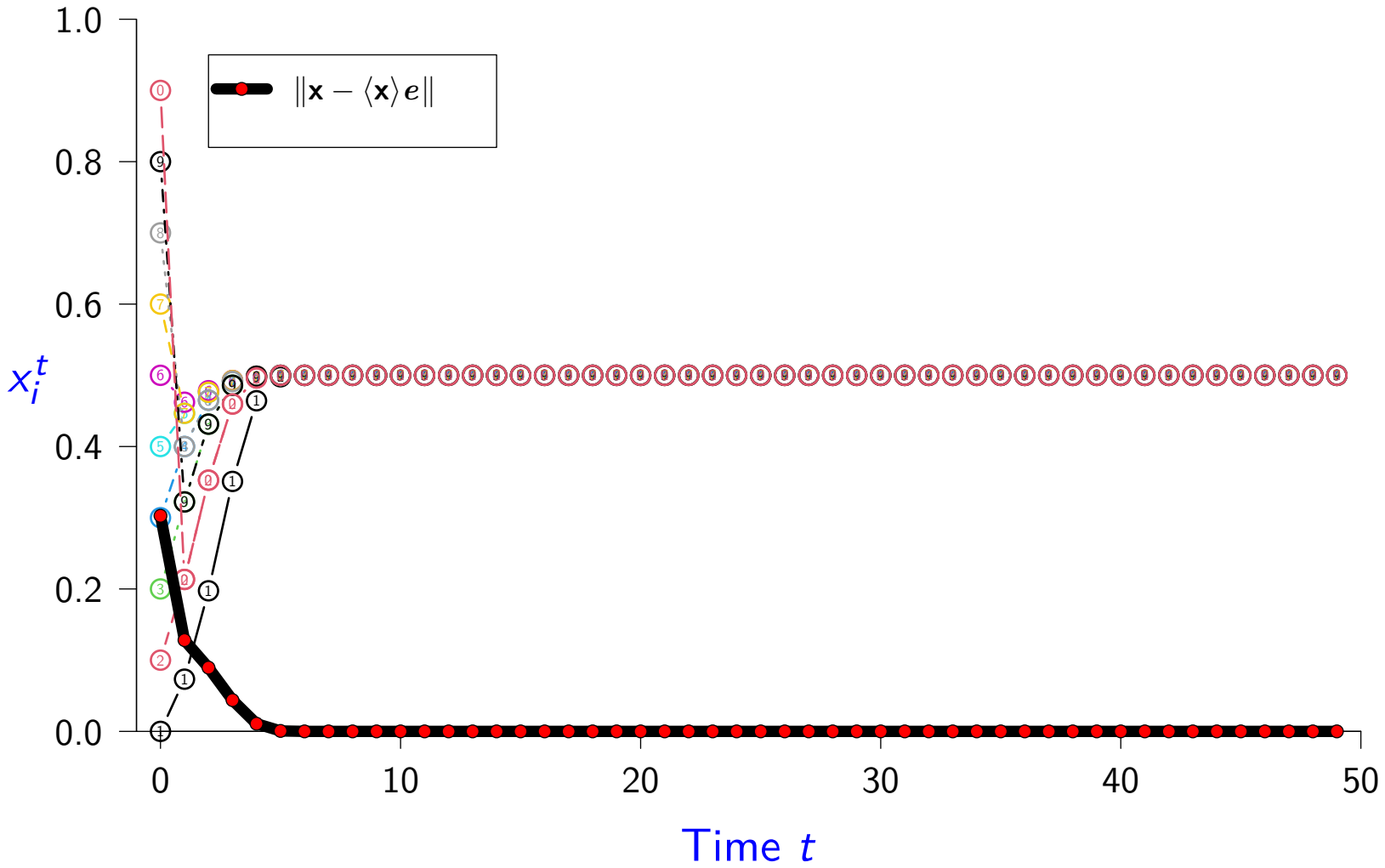


Logistic Metapopulation Simulation ($r = 2,\ m = 0.2$)

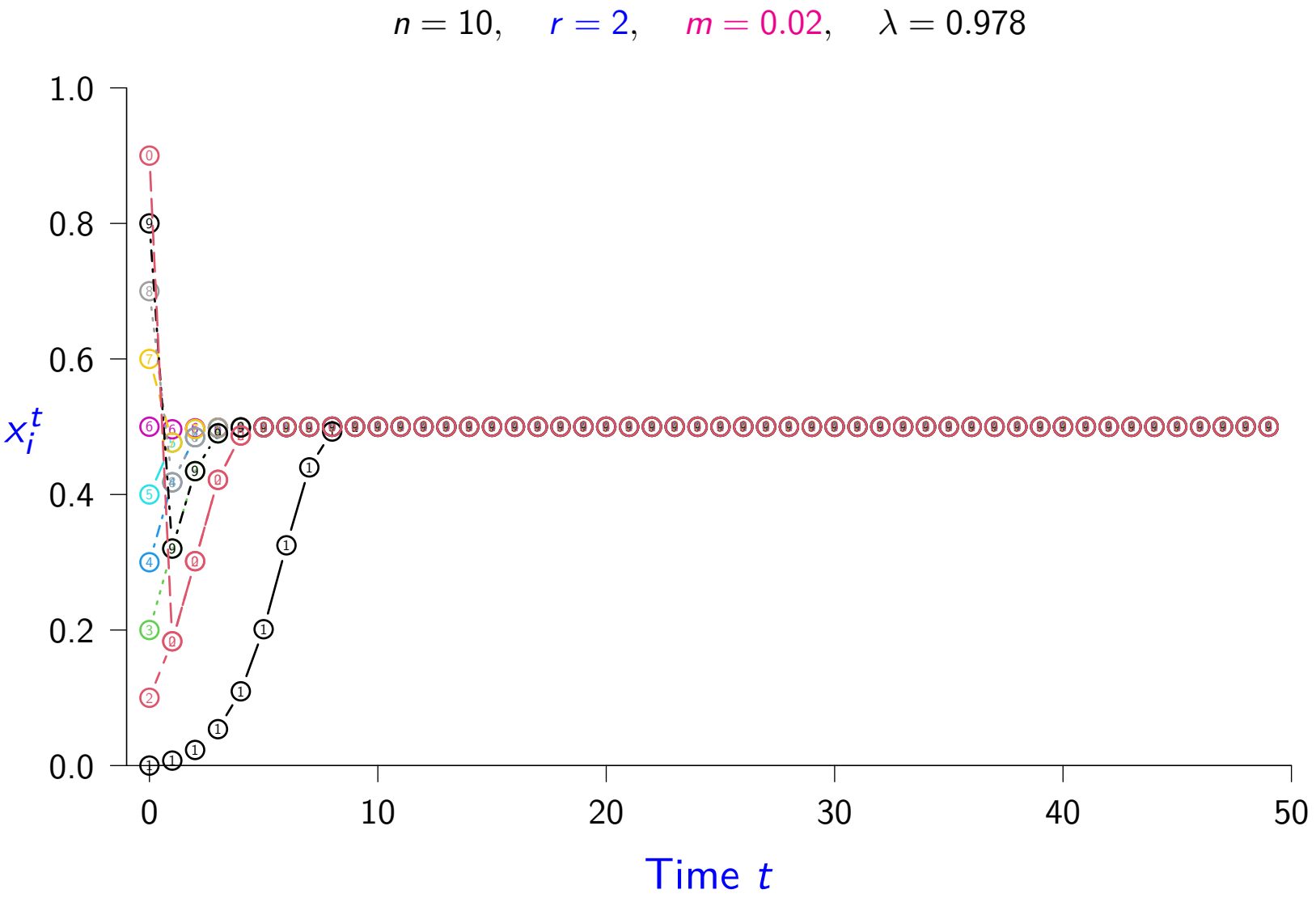


Logistic Metapopulation Simulation ($r = 2,\ m = 0.2$)

$n = 10, \quad r = 2, \quad m = 0.2, \quad \lambda = 0.778$

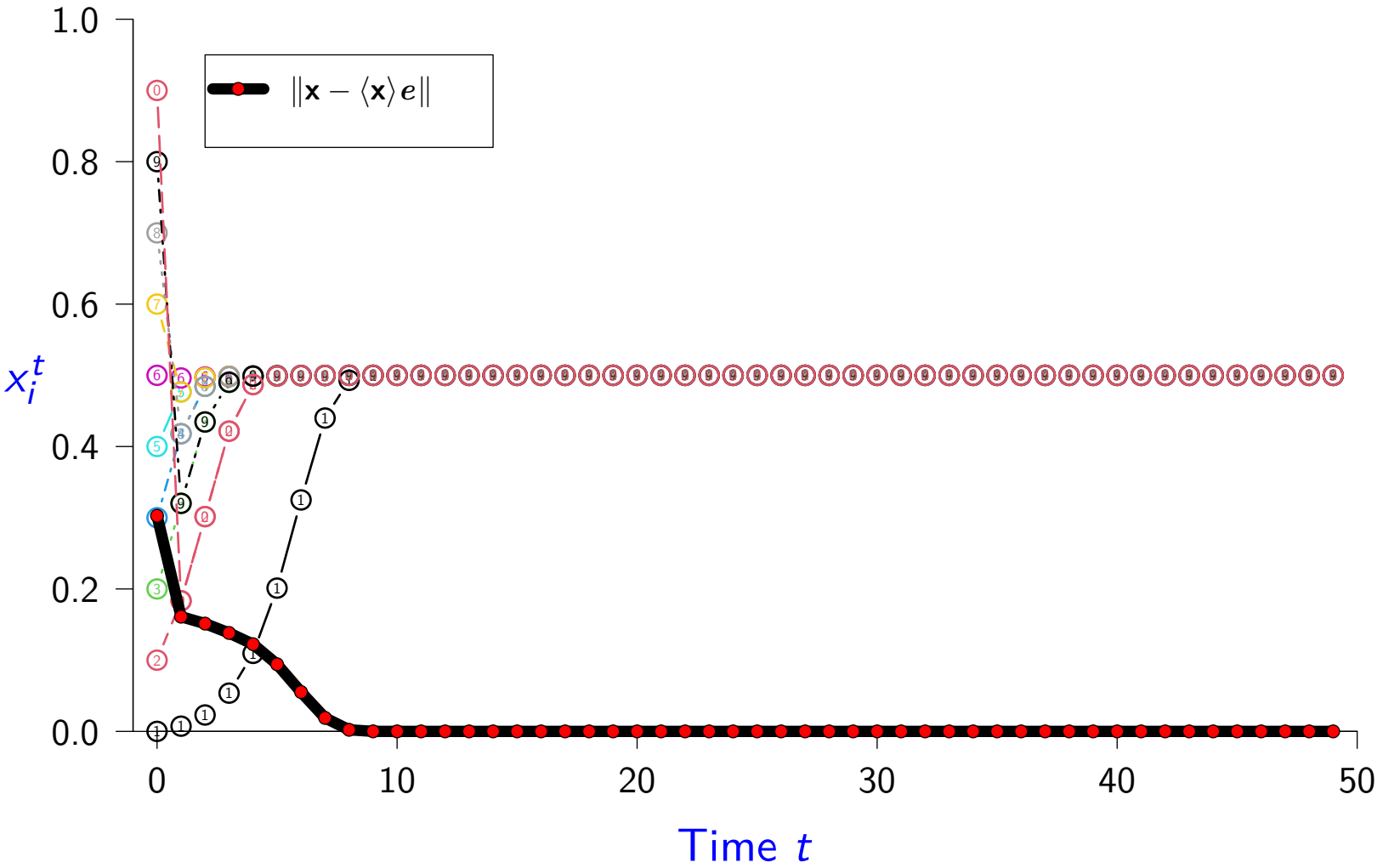


Logistic Metapopulation Simulation ($r = 2,\ m = 0.02$)

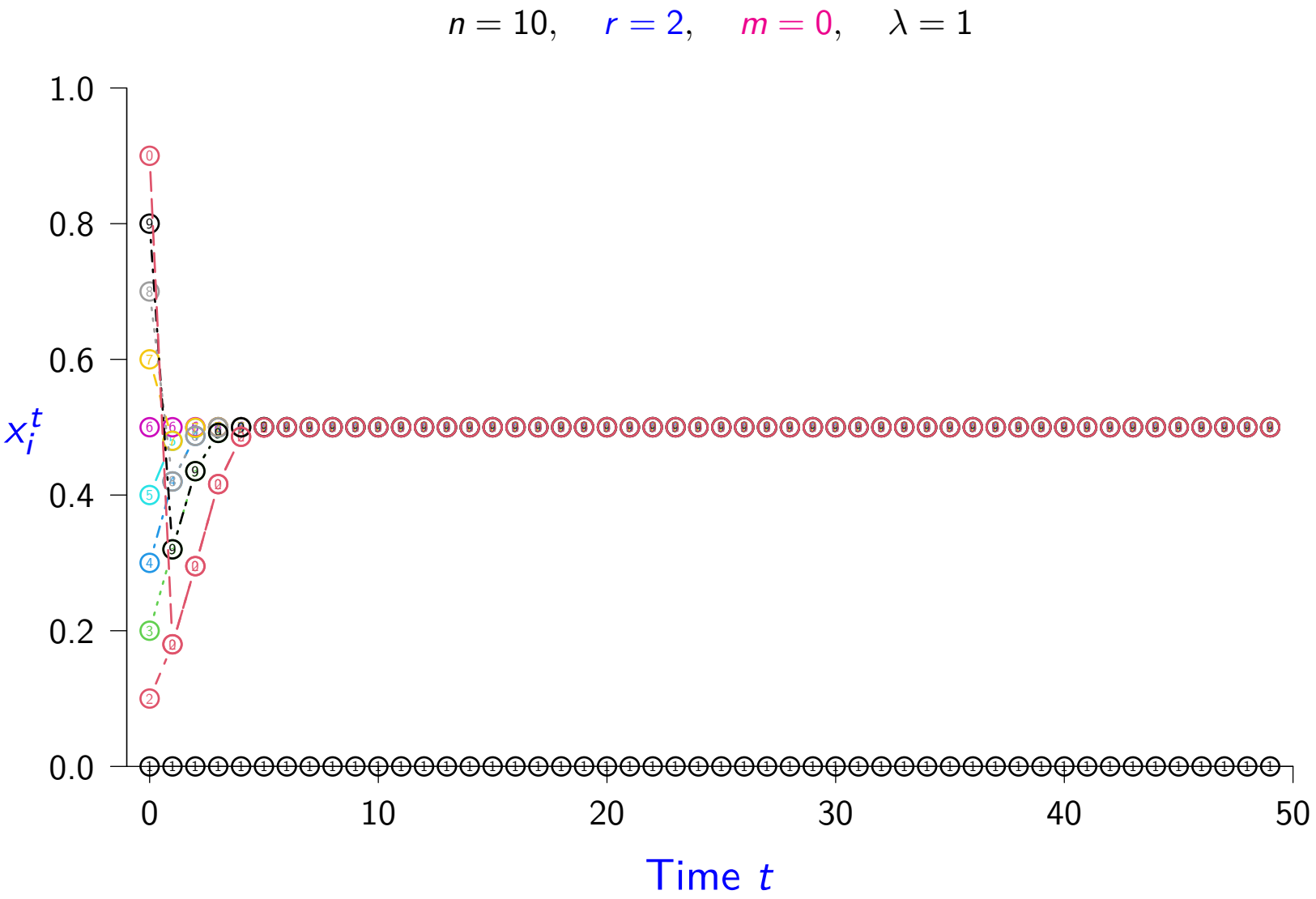


Logistic Metapopulation Simulation ($r = 2,\ m = 0.02$)

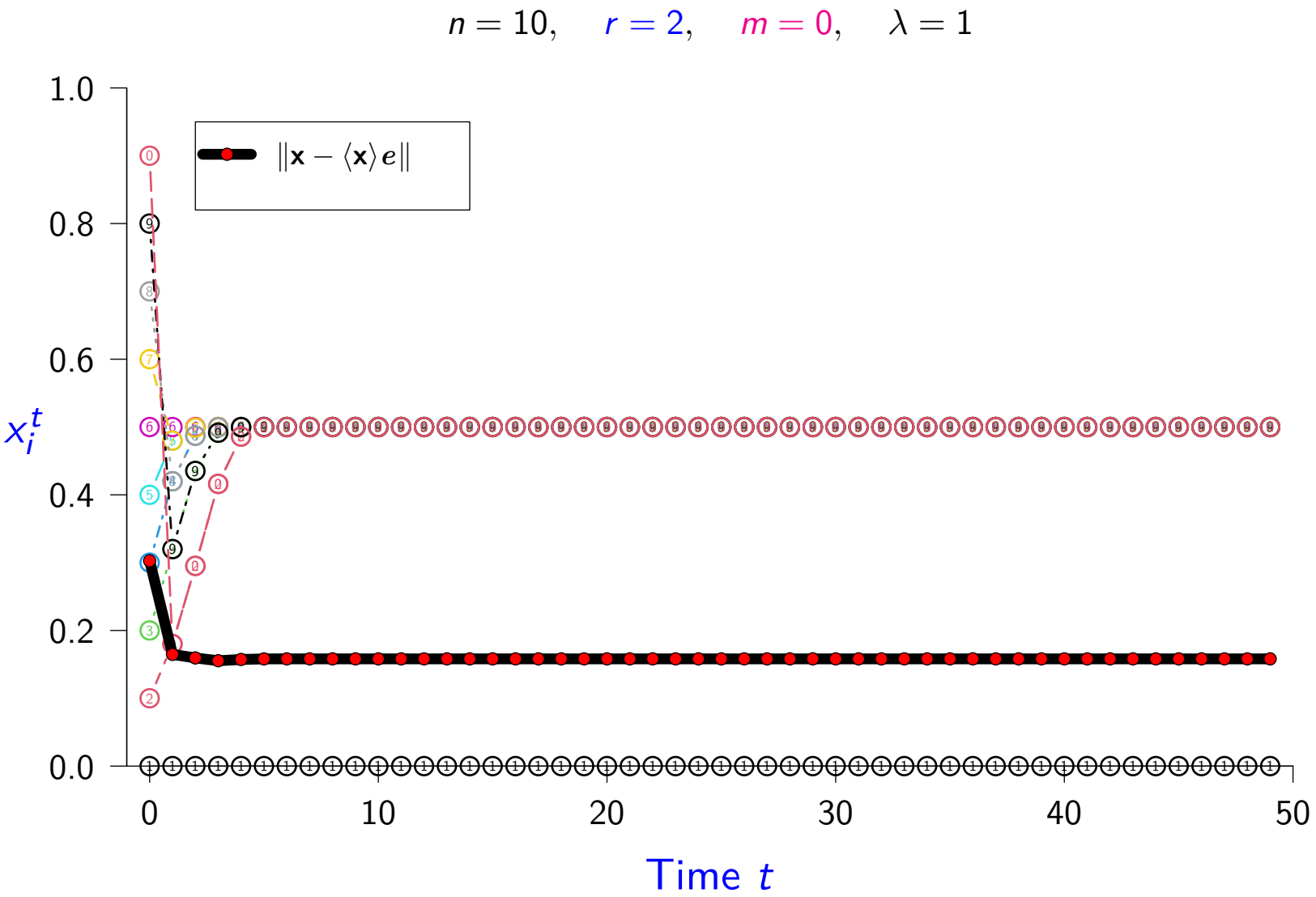
$n = 10, \quad r = 2, \quad m = 0.02, \quad \lambda = 0.978$



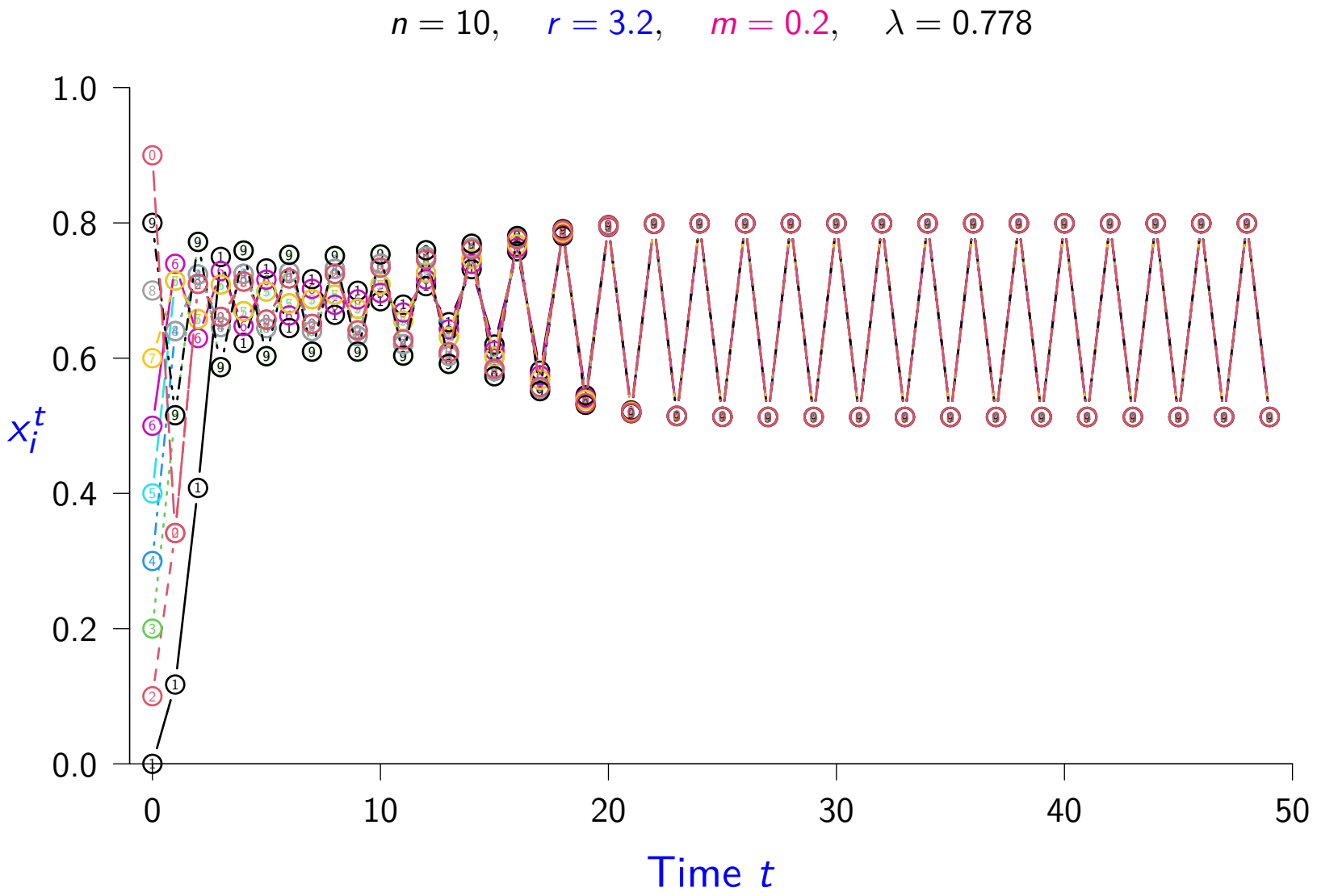
Logistic Metapopulation Simulation ($r = 2,\ m = 0$)



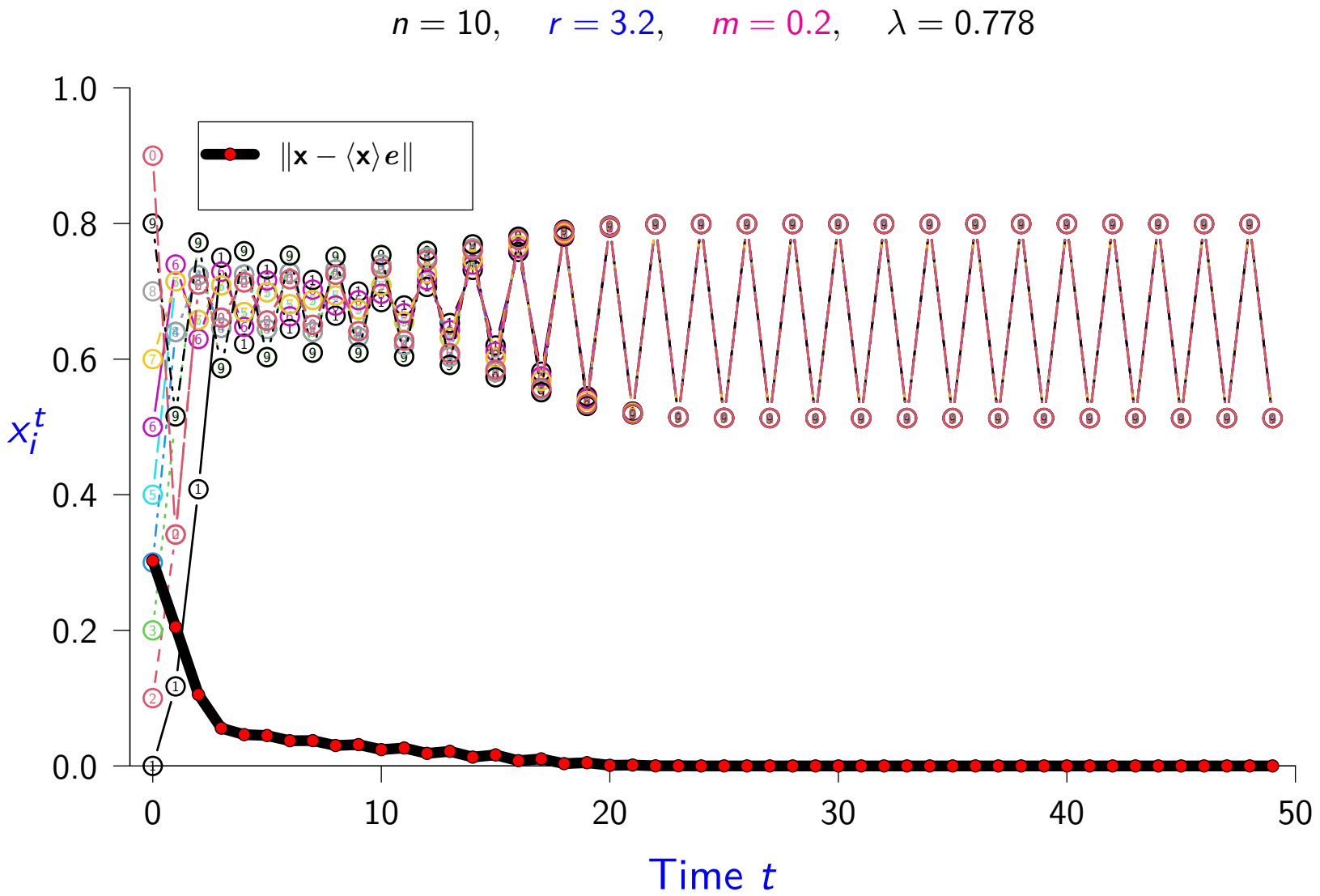
Logistic Metapopulation Simulation ($r = 2,\ m = 0$)



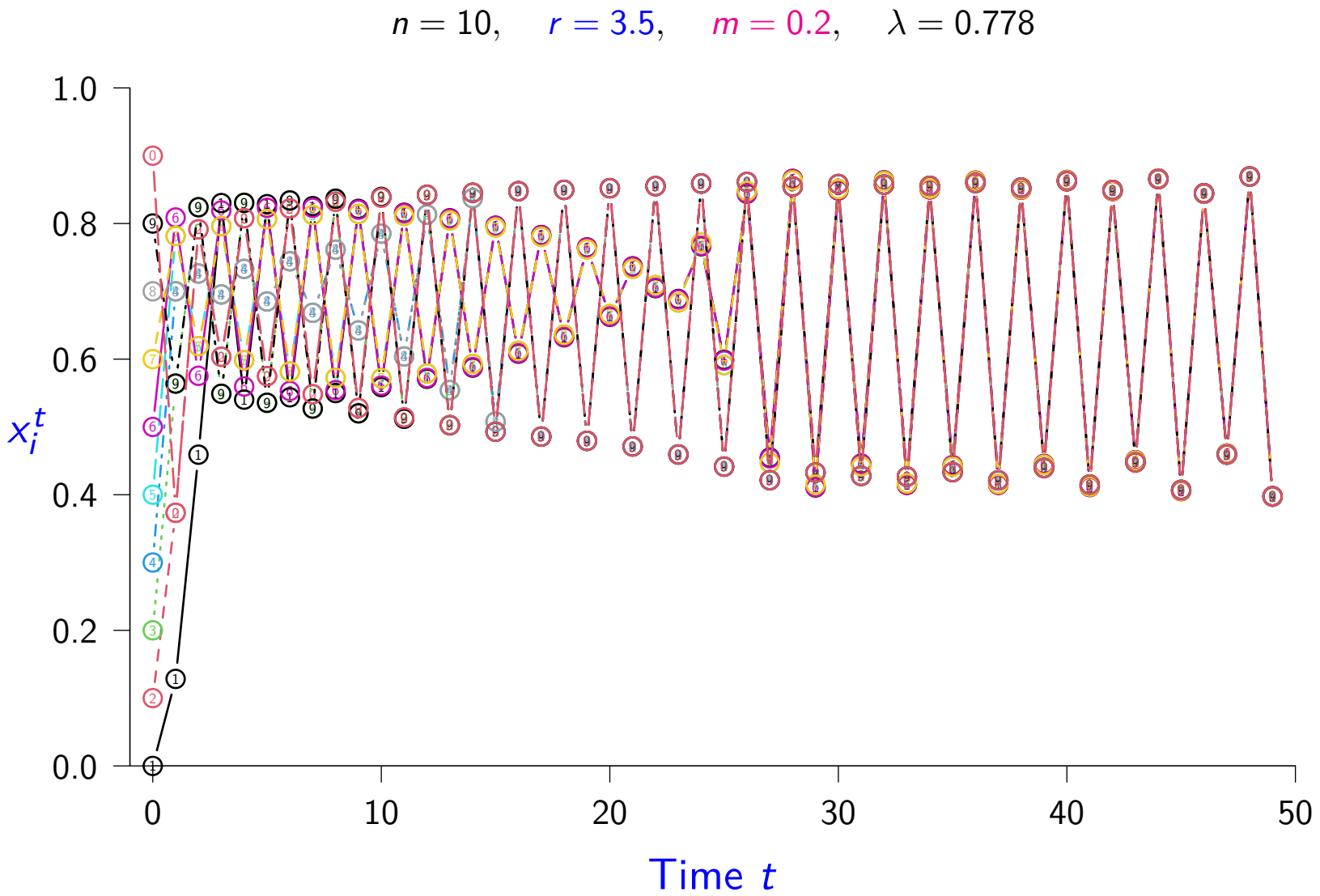
Logistic Metapopulation Simulation ($r = 3.2$, $m = 0.2$)



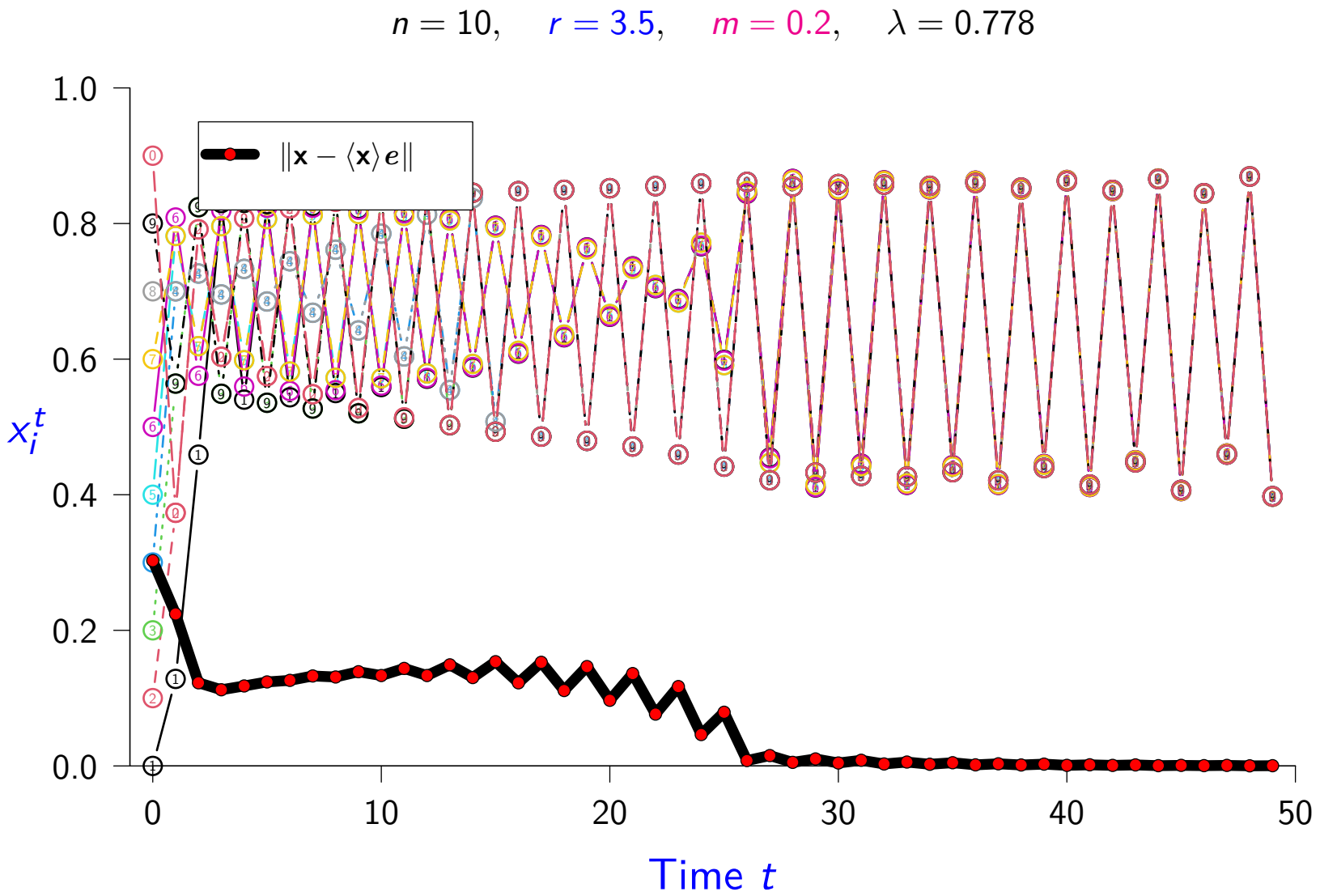
Logistic Metapopulation Simulation ($r = 3.2$, $m = 0.2$)



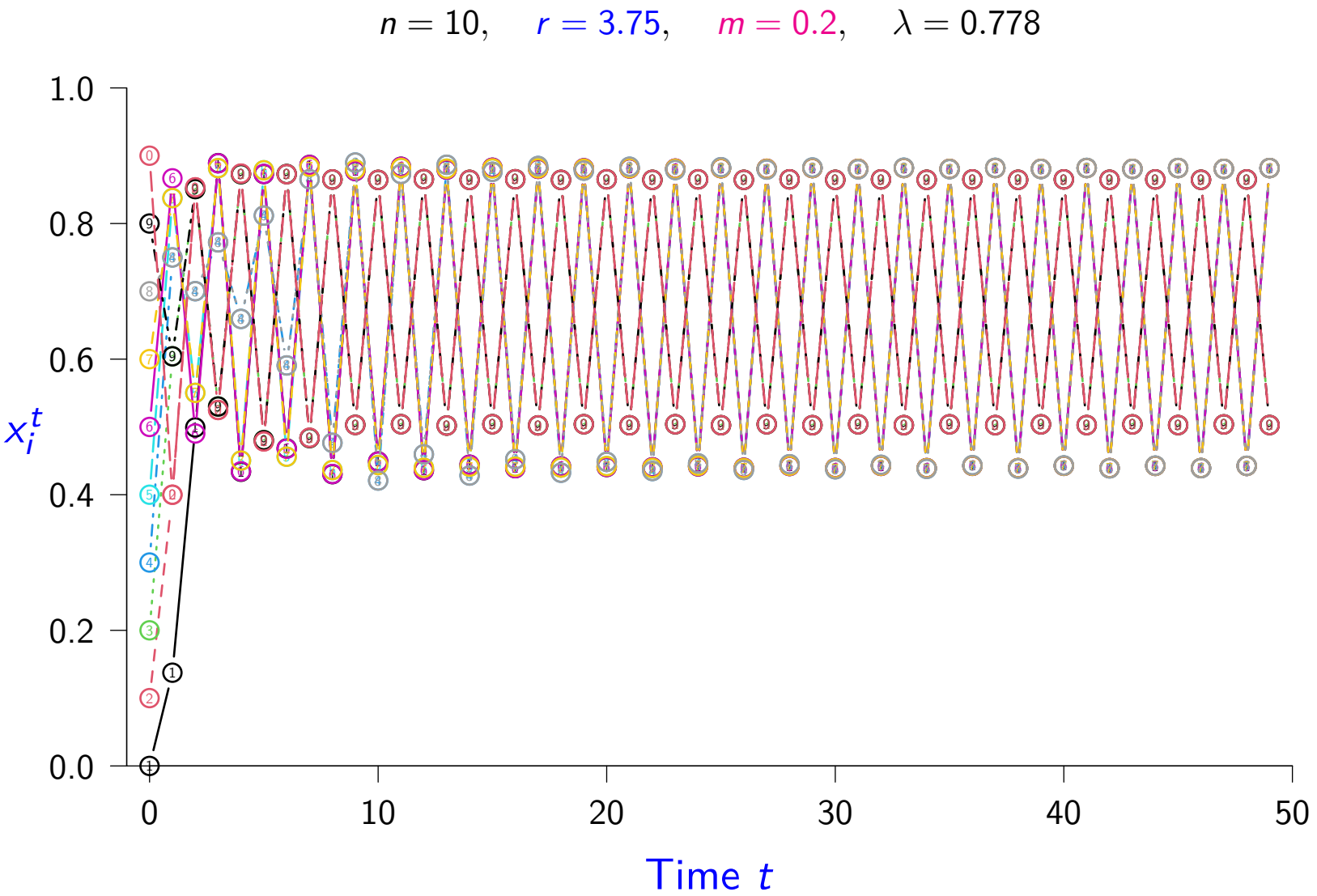
Logistic Metapopulation Simulation ($r = 3.5$, $m = 0.2$)



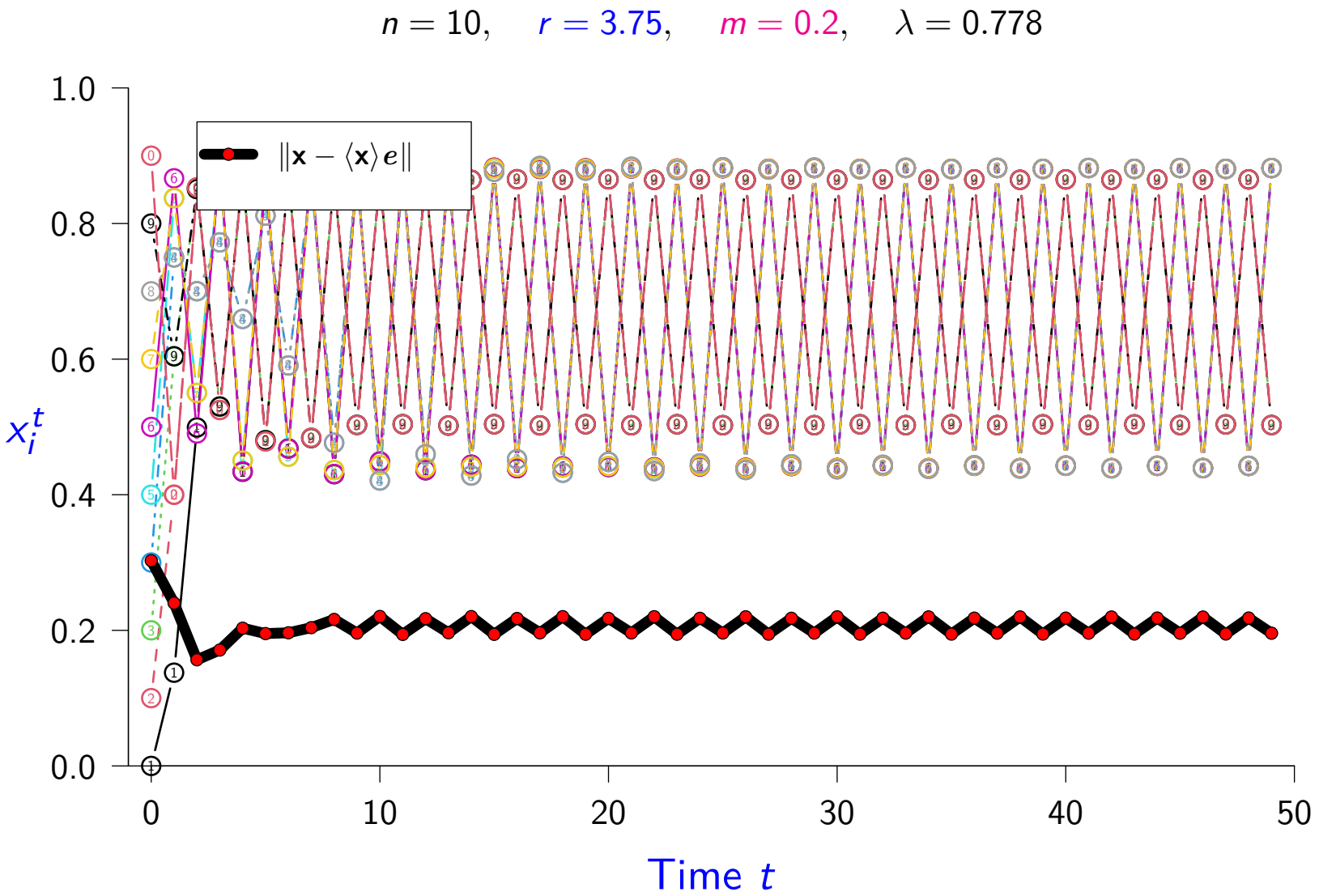
Logistic Metapopulation Simulation ($r = 3.5$, $m = 0.2$)



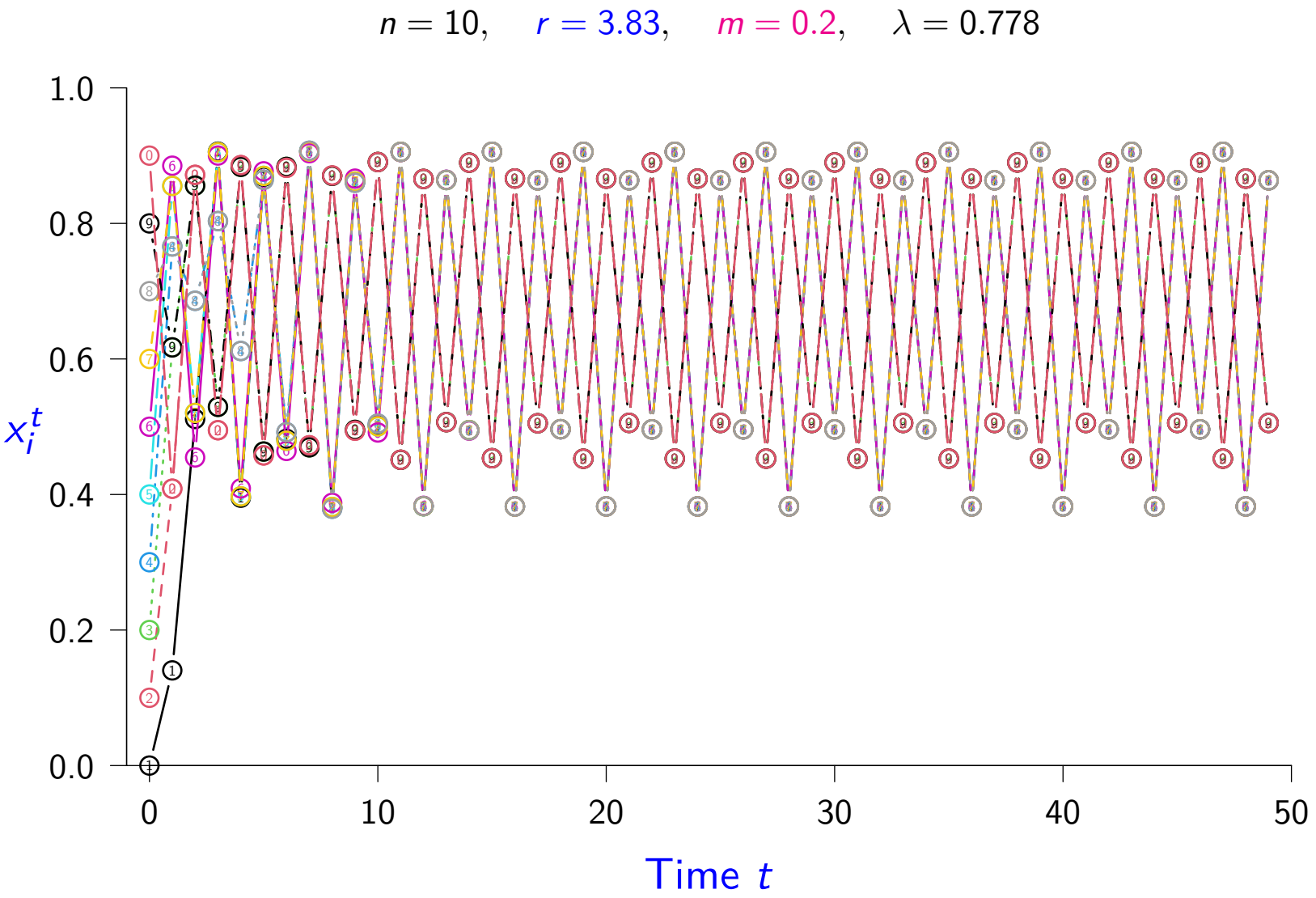
Logistic Metapopulation Simulation ($r = 3.75,\ m = 0.2$)



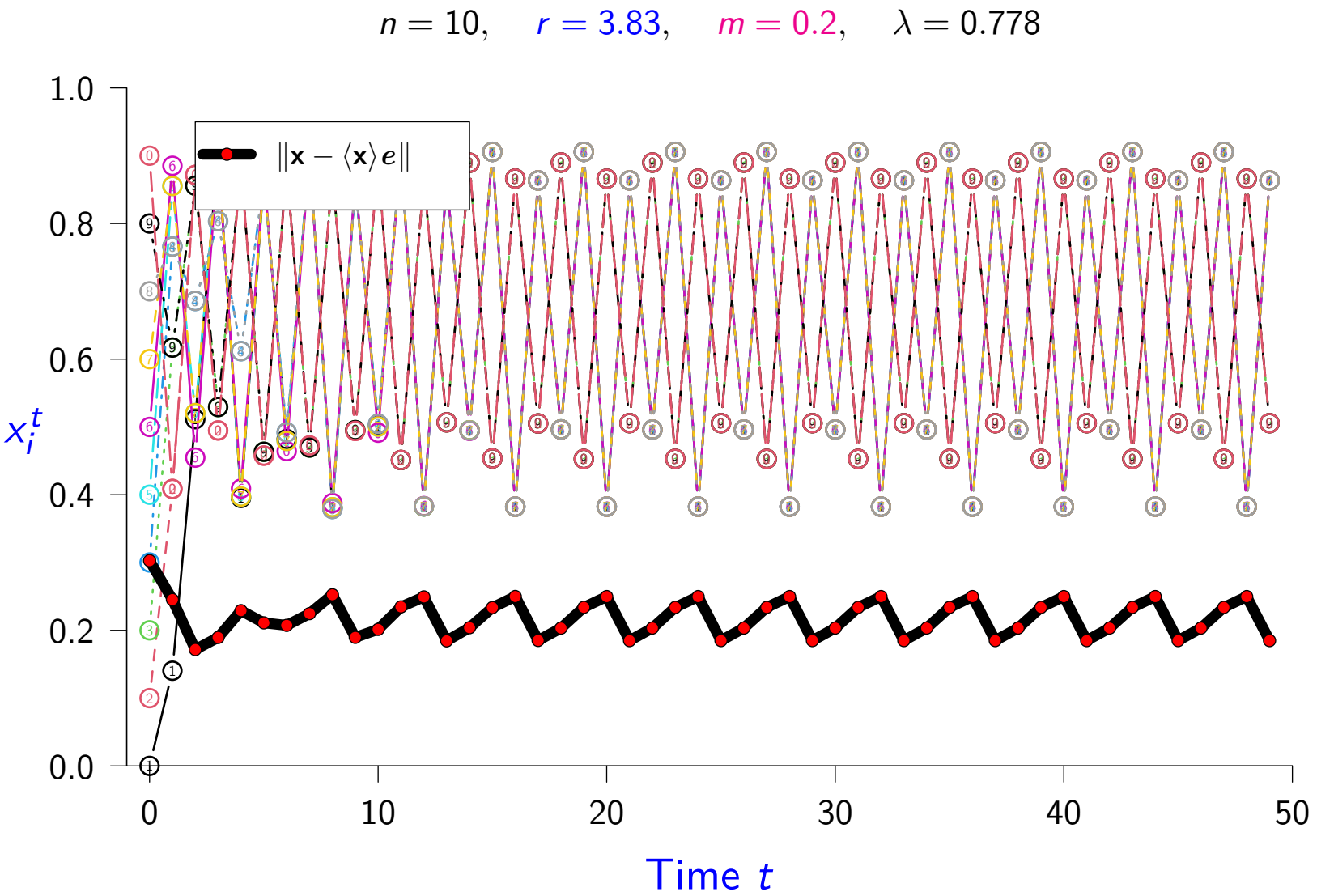
Logistic Metapopulation Simulation ($r = 3.75,\ m = 0.2$)



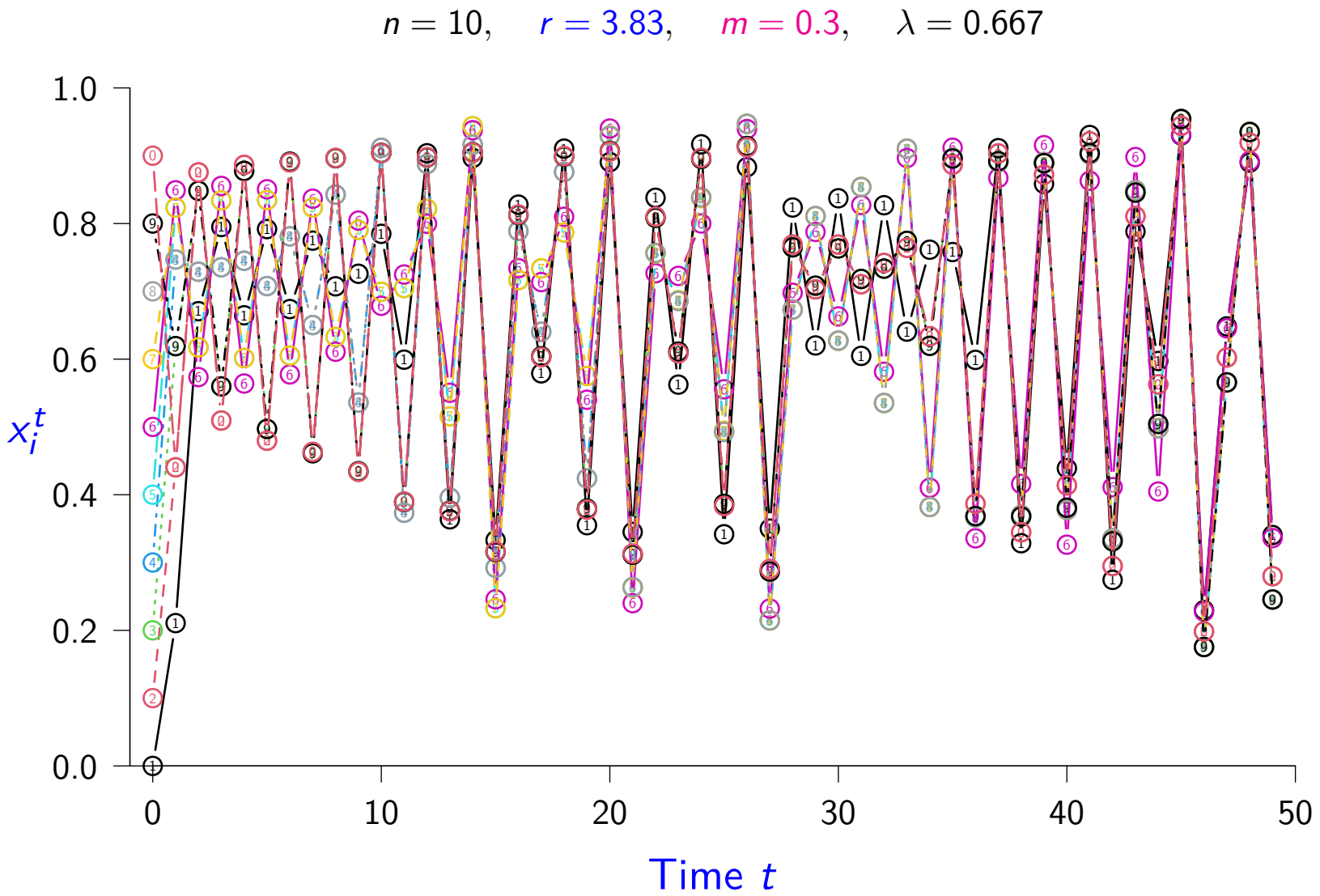
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.2$)



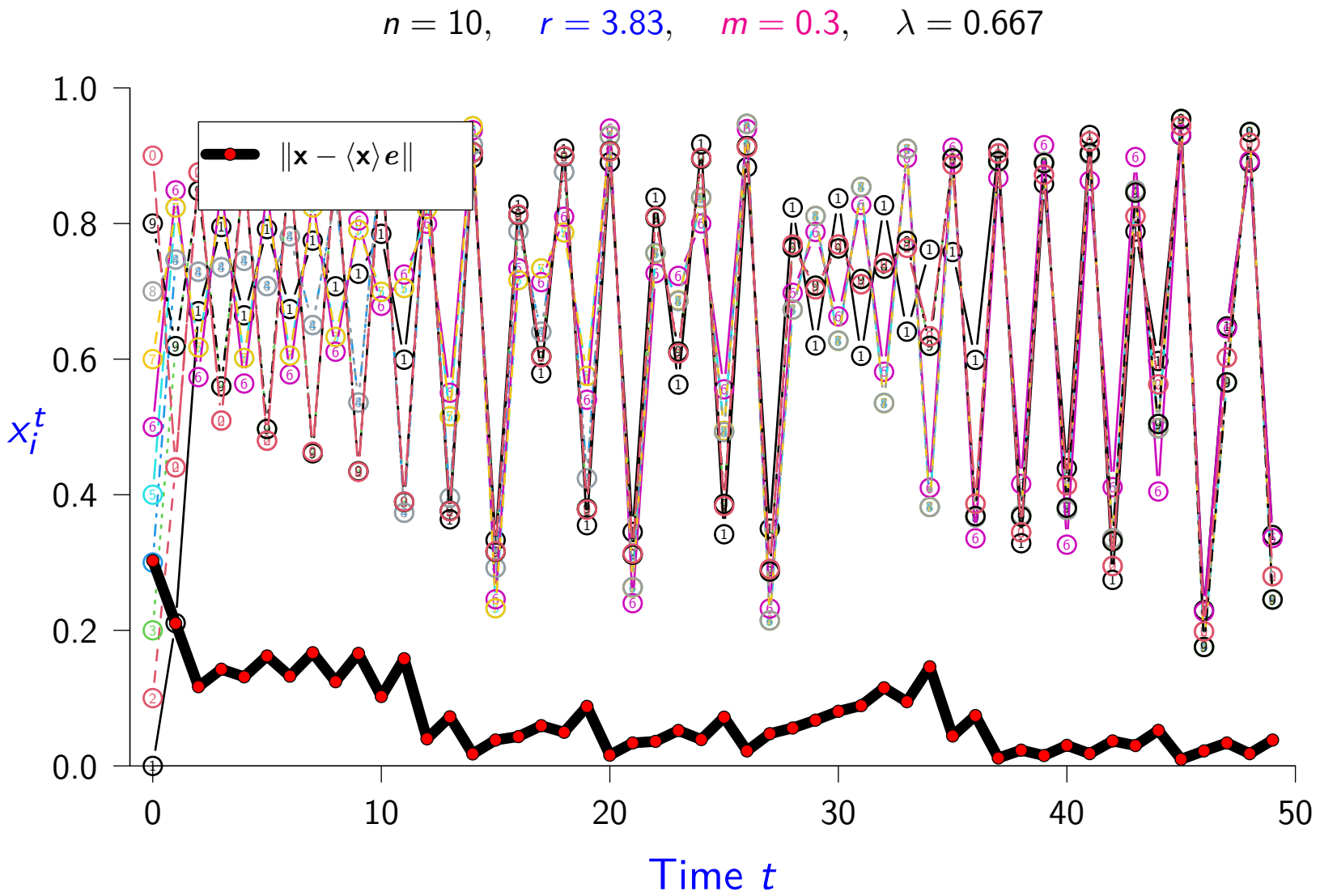
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.2$)



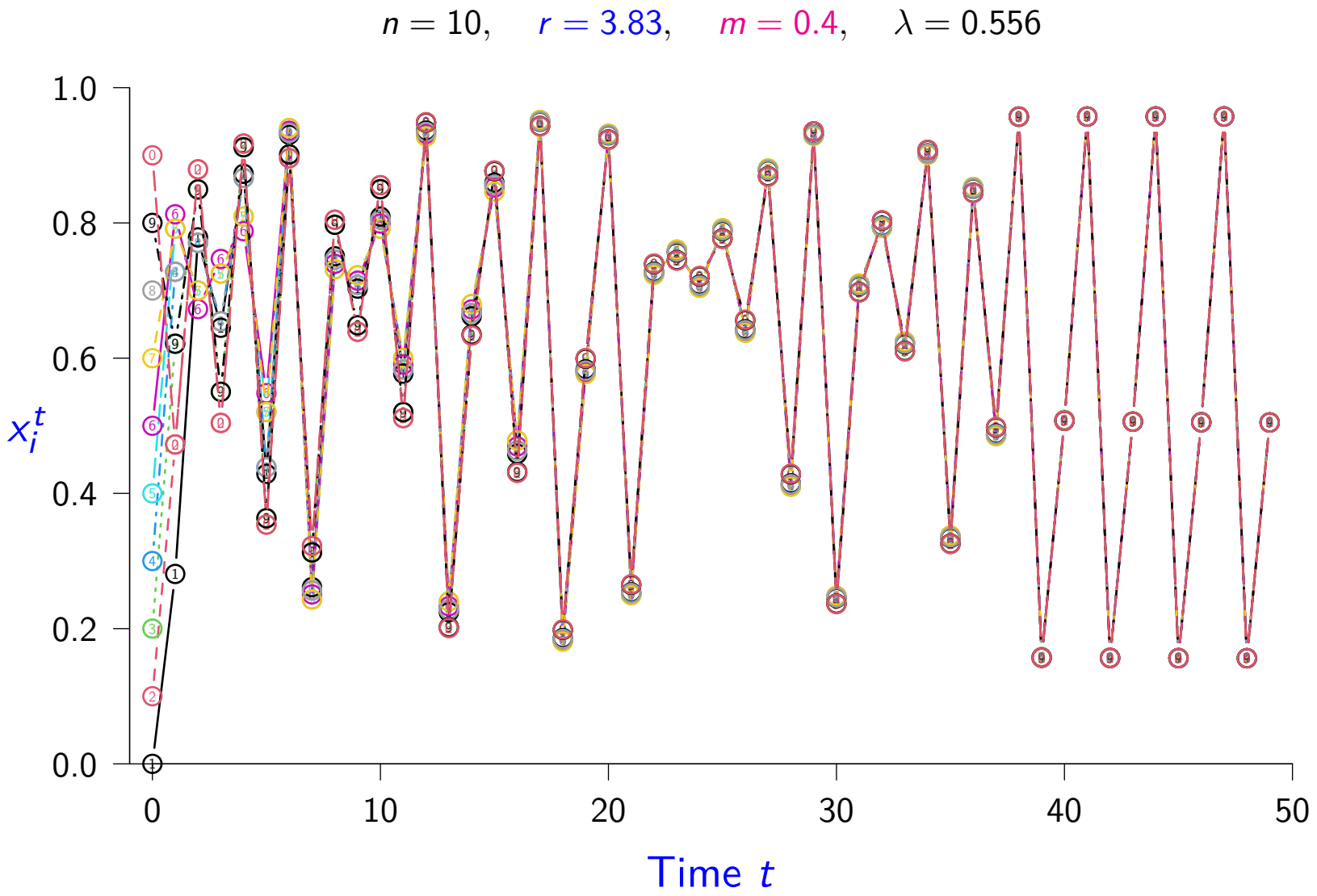
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.3$)



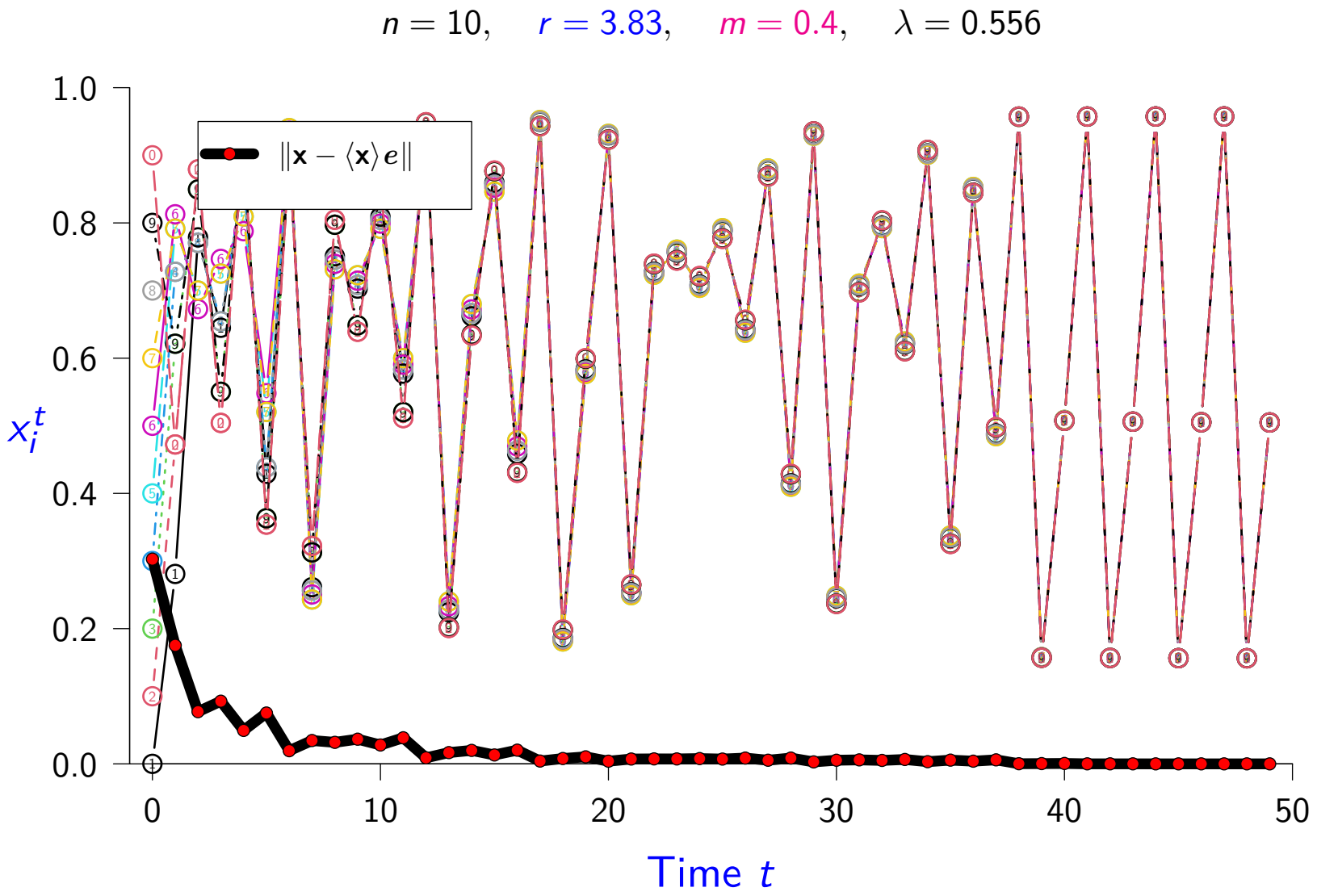
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.3$)



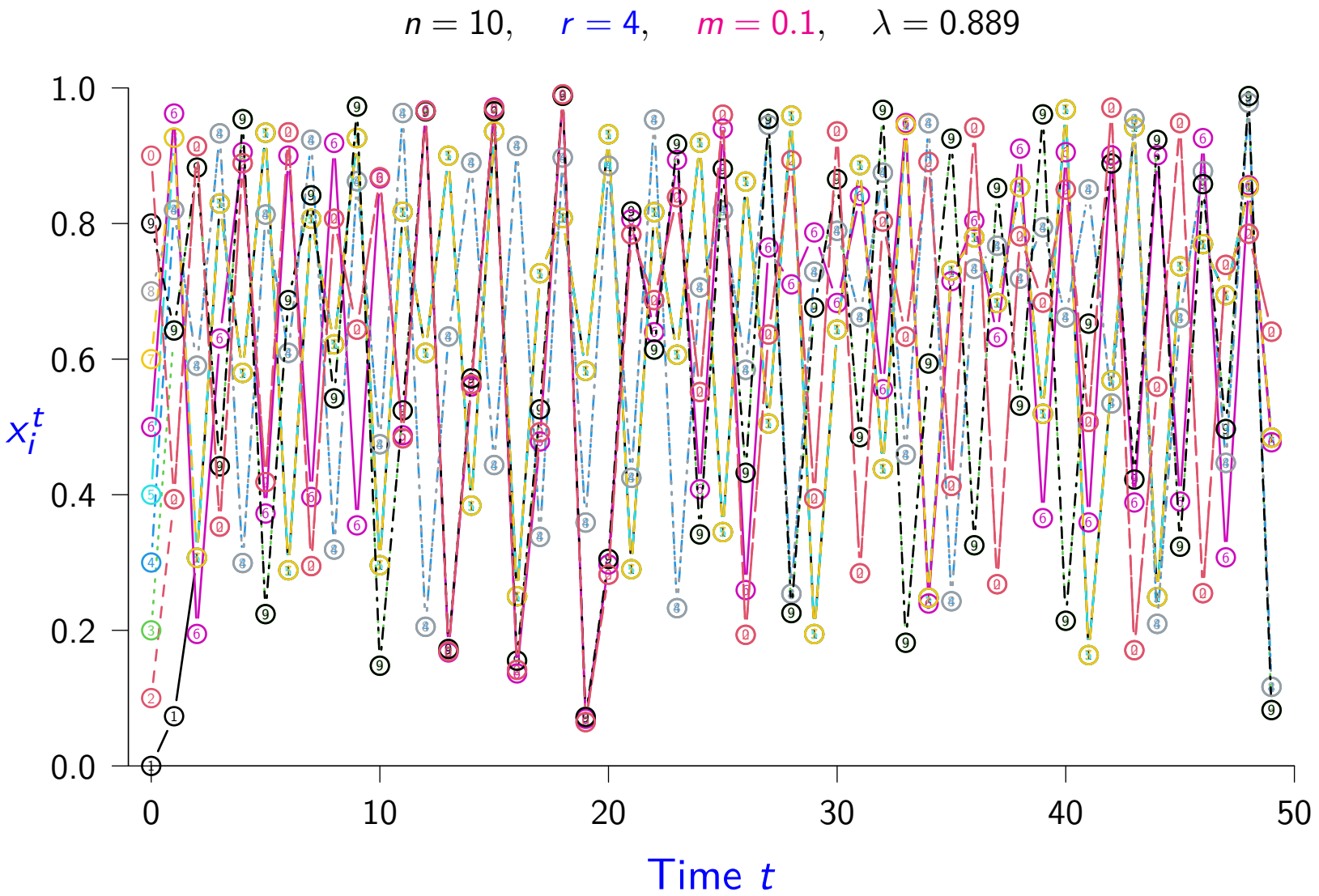
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.4$)



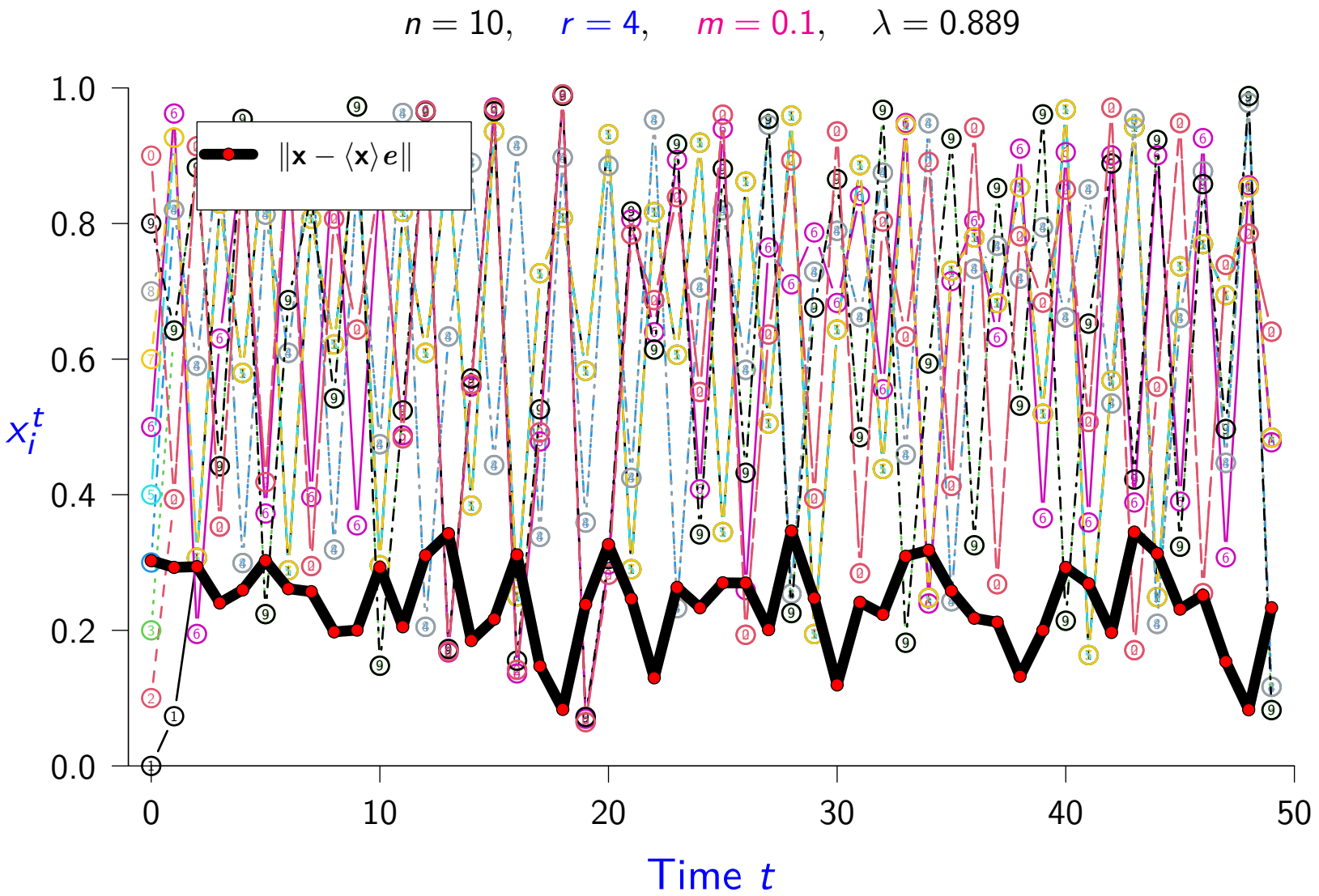
Logistic Metapopulation Simulation ($r = 3.83,\ m = 0.4$)



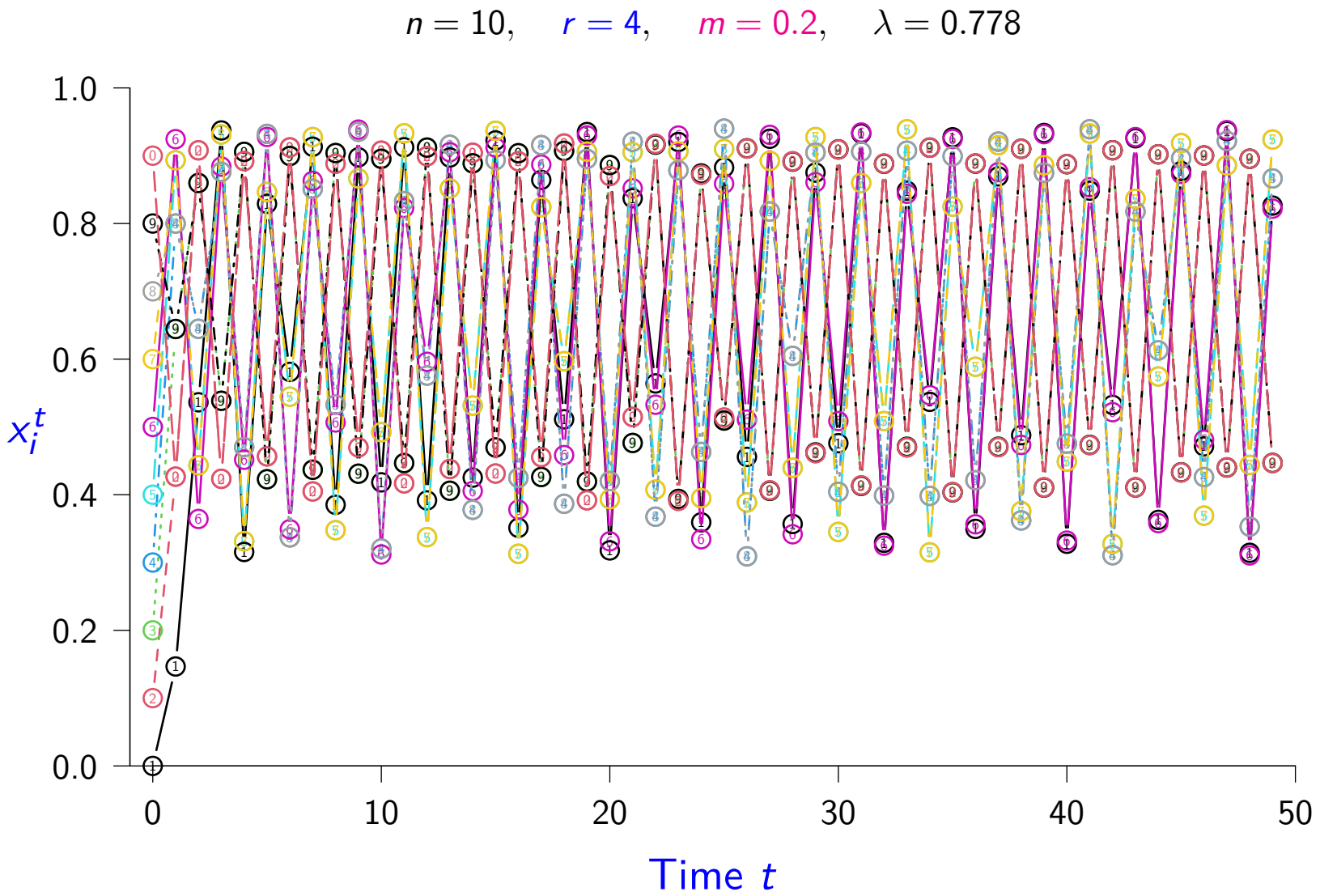
Logistic Metapopulation Simulation ($r = 4$, $m = 0.1$)



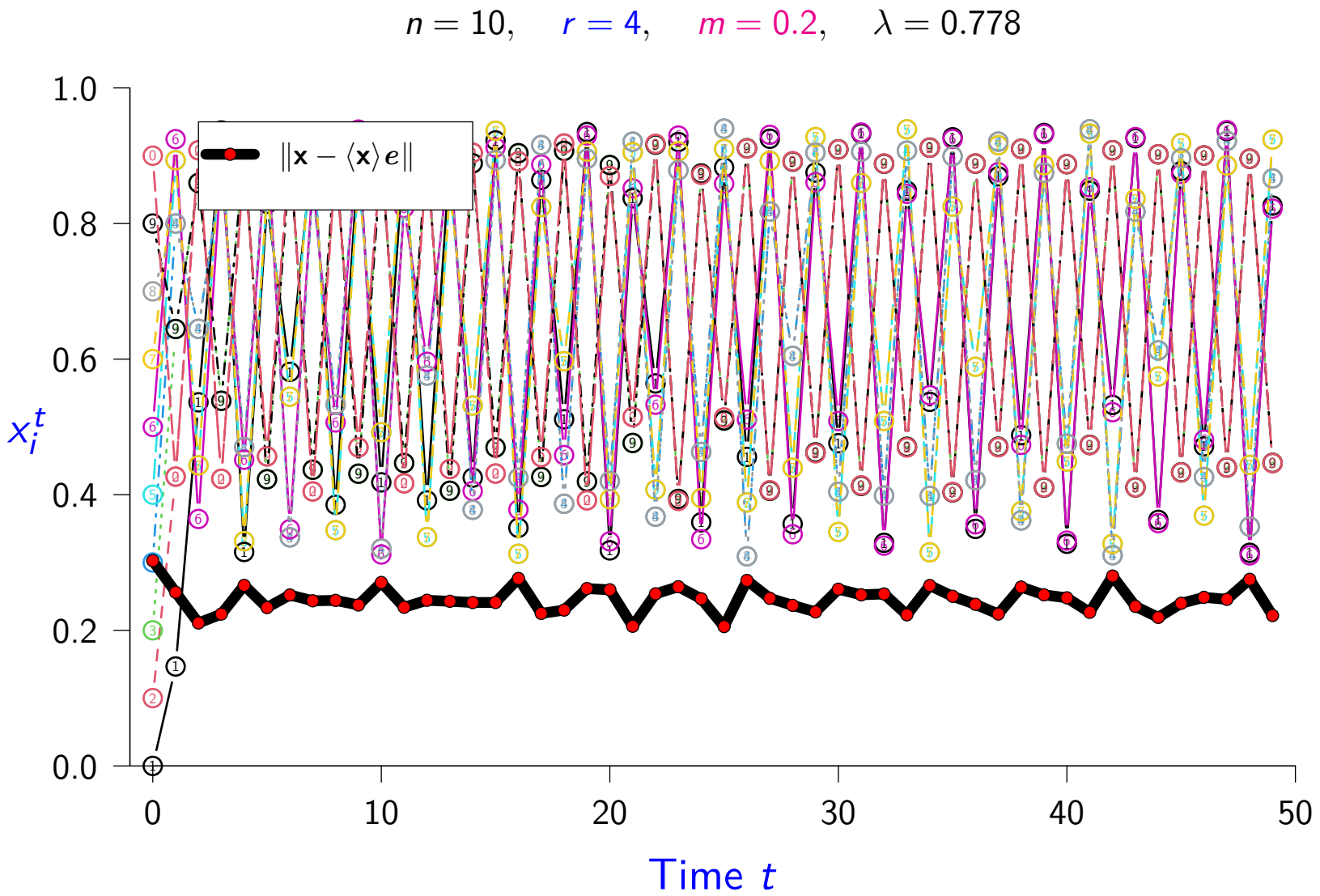
Logistic Metapopulation Simulation ($r = 4$, $m = 0.1$)



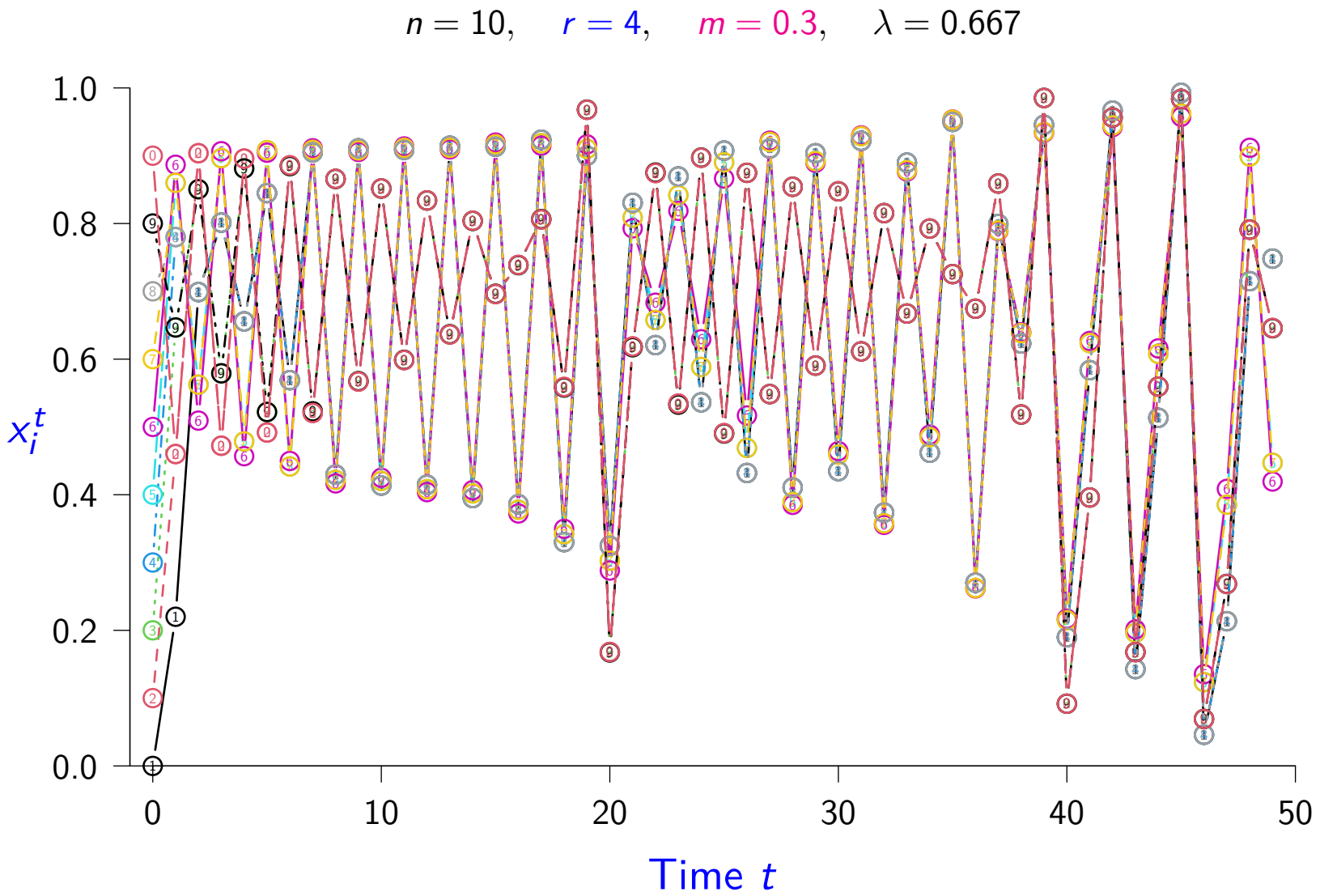
Logistic Metapopulation Simulation ($r = 4$, $m = 0.2$)



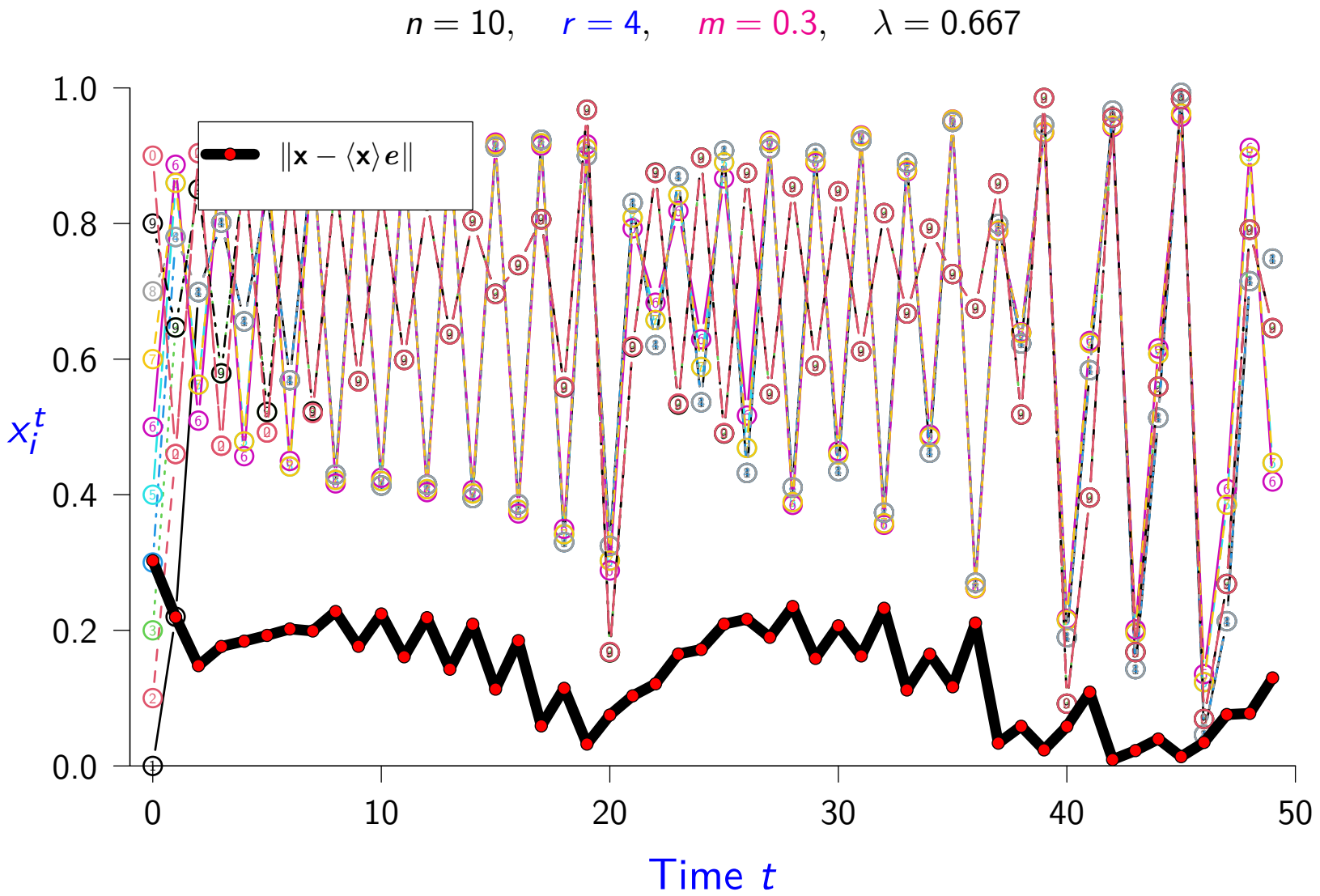
Logistic Metapopulation Simulation ($r = 4$, $m = 0.2$)



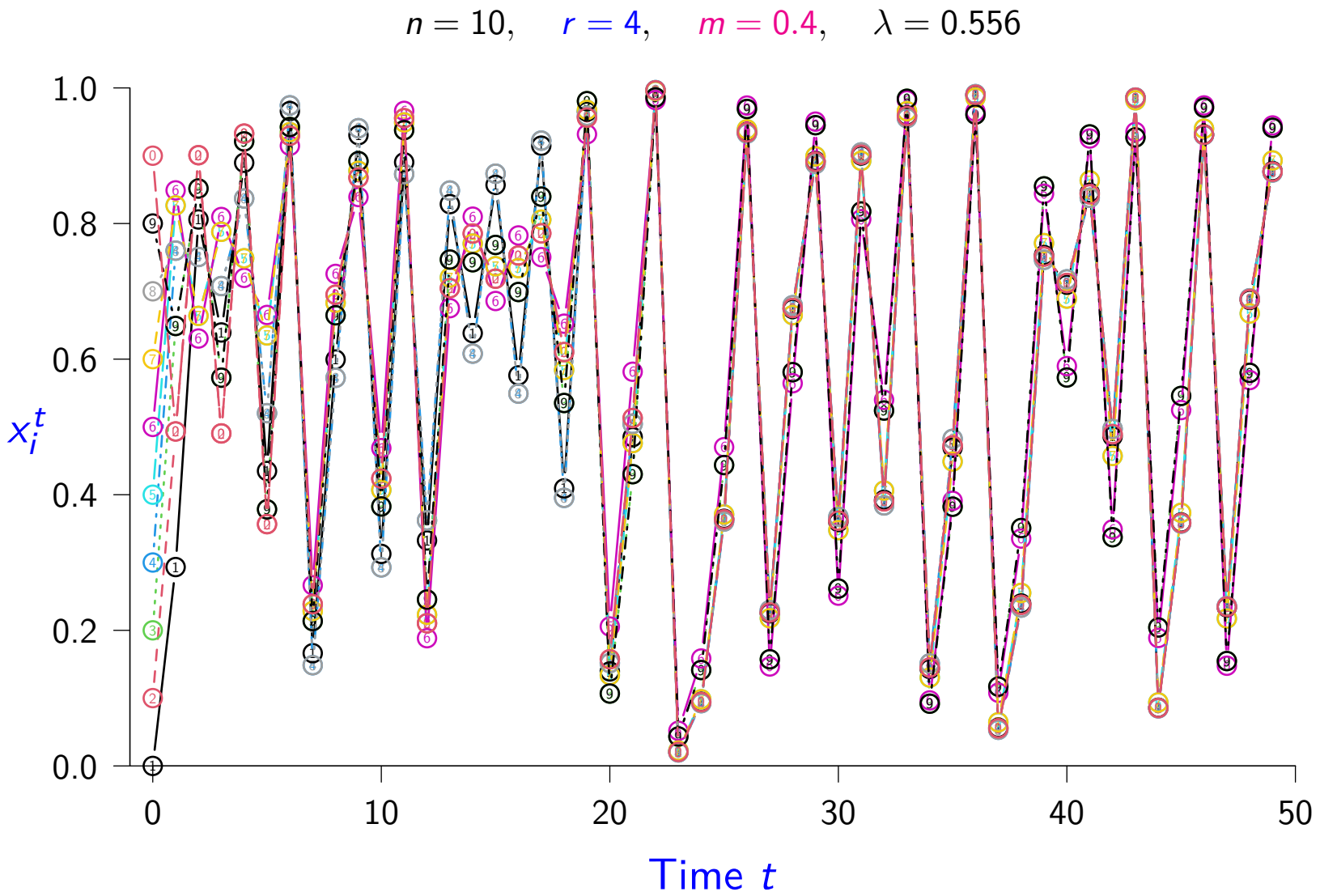
Logistic Metapopulation Simulation ($r = 4$, $m = 0.3$)



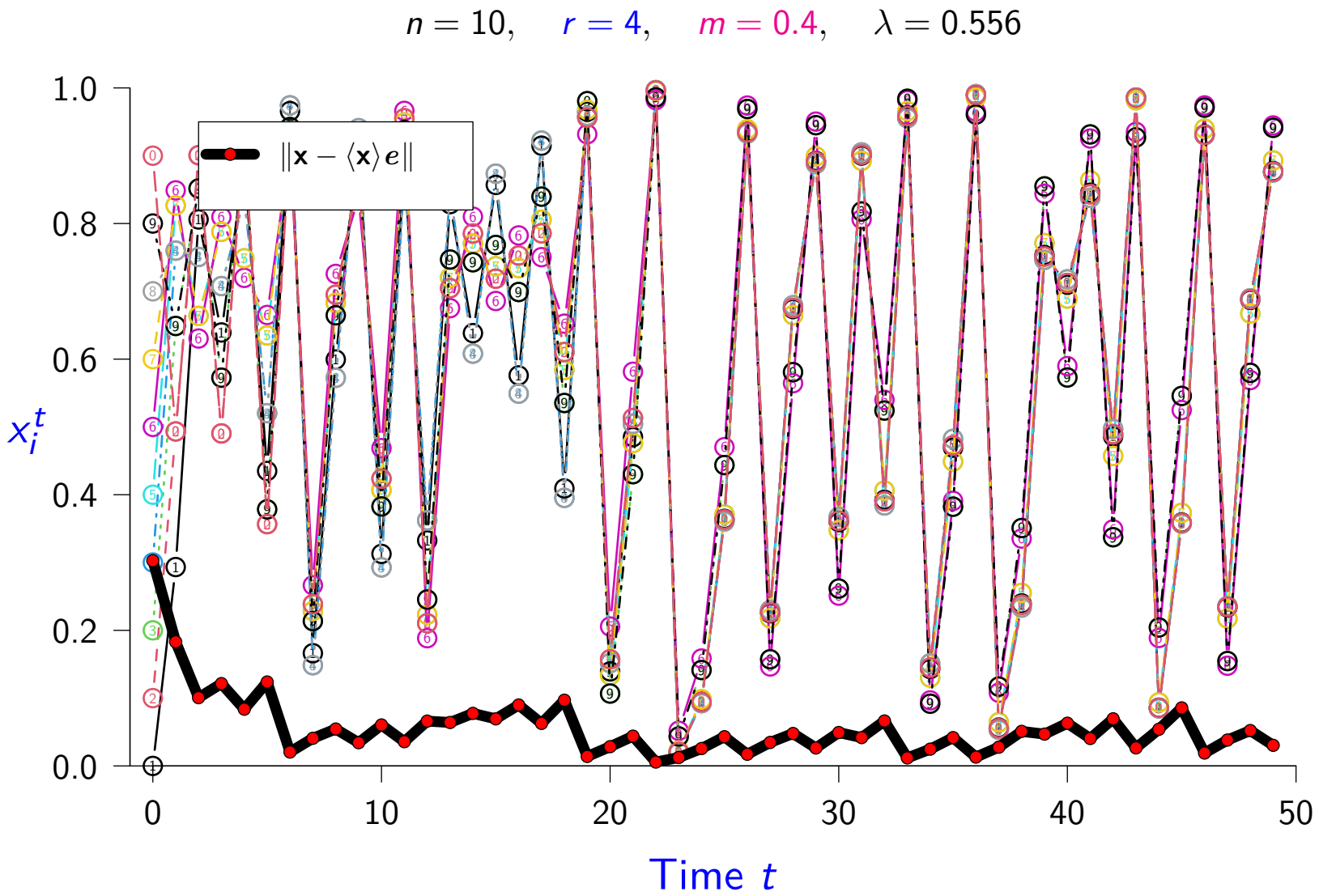
Logistic Metapopulation Simulation ($r = 4$, $m = 0.3$)



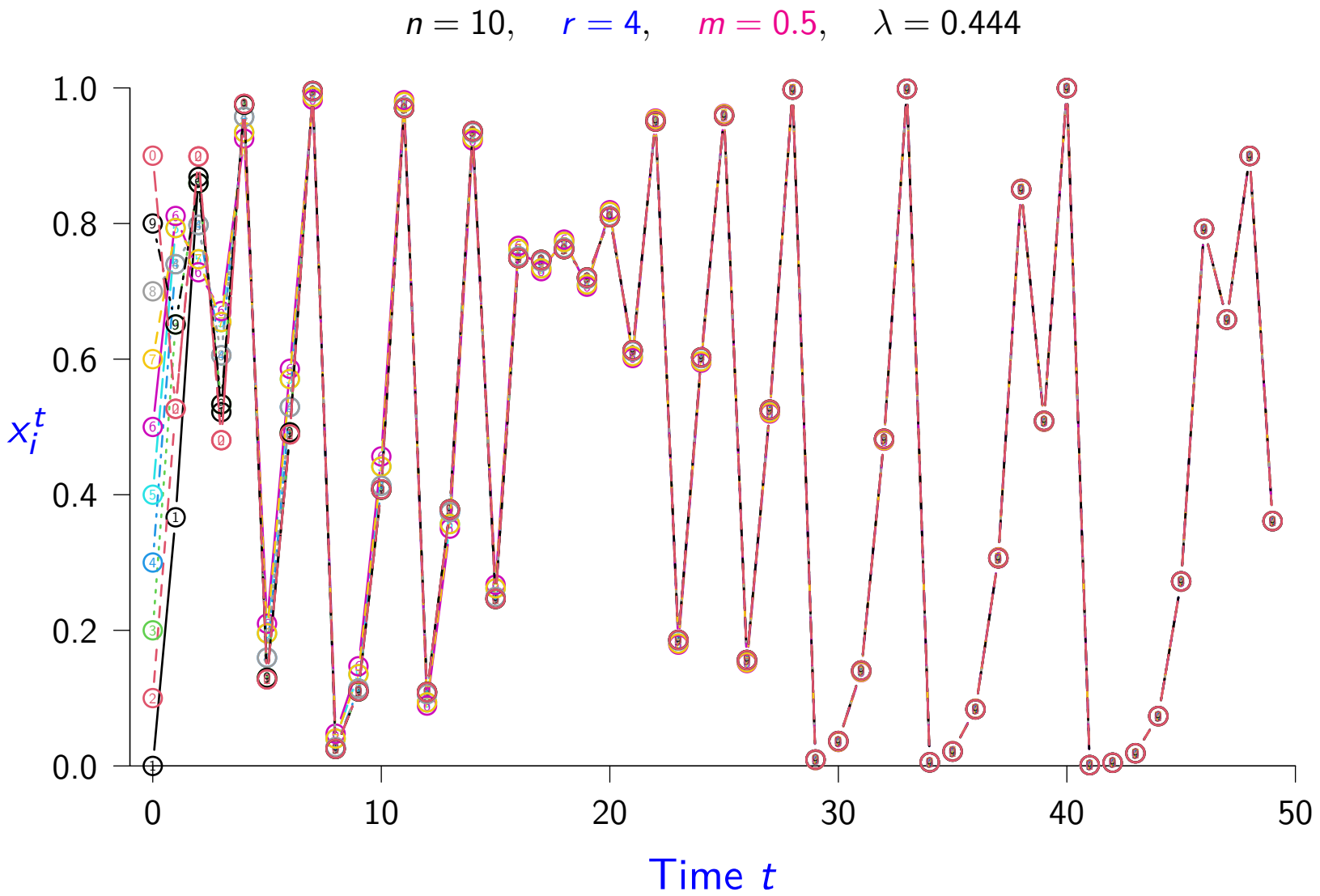
Logistic Metapopulation Simulation ($r = 4$, $m = 0.4$)



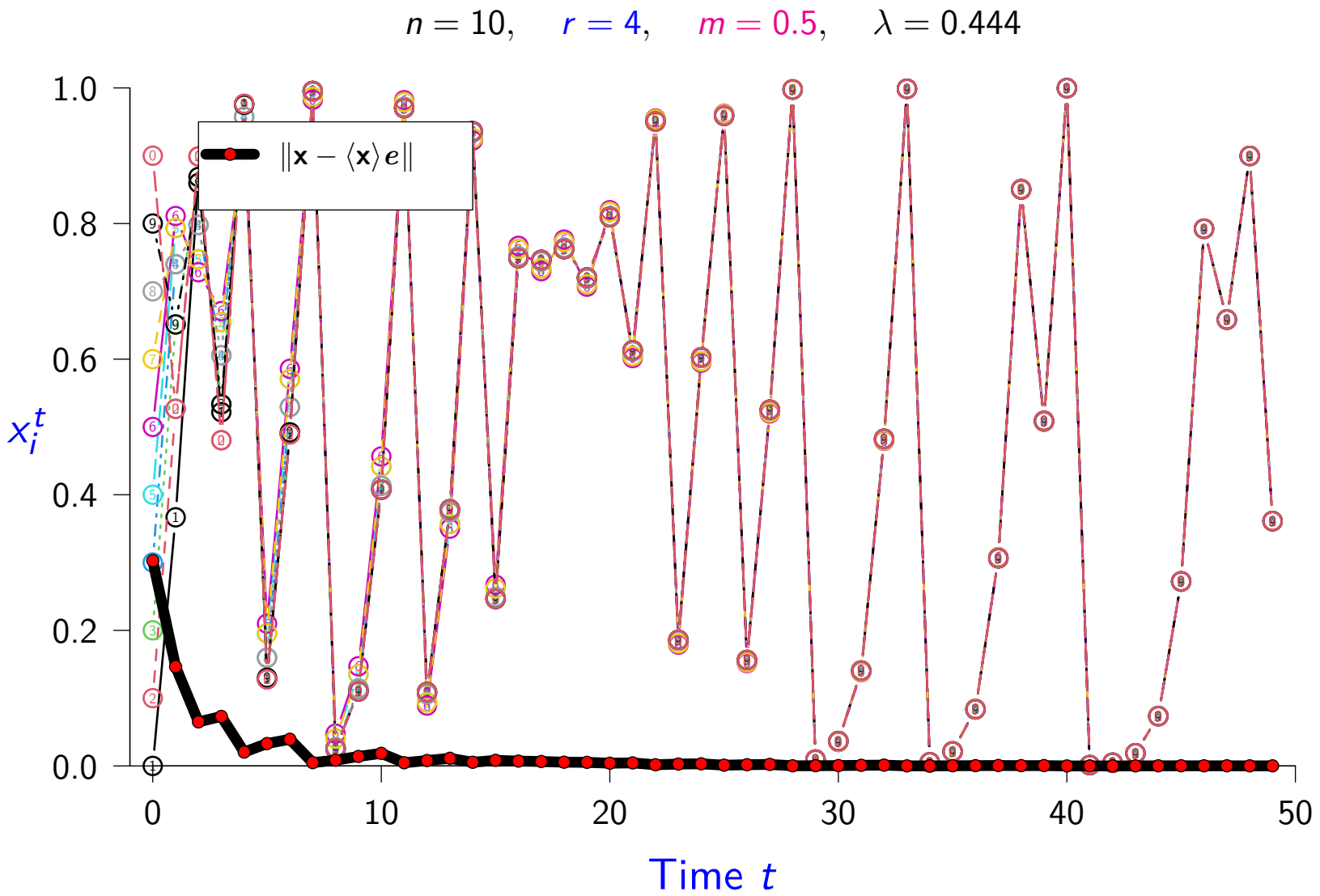
Logistic Metapopulation Simulation ($r = 4$, $m = 0.4$)



Logistic Metapopulation Simulation ($r = 4$, $m = 0.5$)



Logistic Metapopulation Simulation ($r = 4$, $m = 0.5$)



Metapopulation dynamics: what we've seen so far

- Examples of connectivity matrices
  - equal coupling
  - nearest-neighbour coupling on a ring
- Logistic Metapopulation Simulations (10 patches)

- $r = 1,\ m = 0.2$
- $r = 2,\ m = 0.2$
- $r = 2,\ m = 0.02$
- $r = 2,\ m = 0$
- $r = 3.2,\ m = 0.2$
- $r = 3.5,\ m = 0.2$
- $r = 3.75,\ m = 0.2$
- $r = 3.83,\ m = 0.2$
- $r = 3.83,\ m = 0.3$
- $r = 3.83,\ m = 0.4$
- $r = 4,\ m = 0.1$
- $r = 4,\ m = 0.2$
- $r = 4,\ m = 0.3$
- $r = 4,\ m = 0.4$
- $r = 4,\ m = 0.5$

Quantities that affect coherence

Degree of spatial coupling:

- Determined by dispersal matrix $M = (m_{ij})$.
- Do we need to worry about all matrix entries? n^2 parameters?
- Are eigenvalues enough?
- Dominant eigenvalue is always 1. Why?
  - Next slide...
- Coherence is affected by magnitude $|\lambda|$ of *subdominant eigenvalue* λ.

Dominant eigenvalue of dispersal matrix M is always 1

Definition (Positive vector)

A vector is ***positive*** if each of its components is positive.

Definition (Dominant eigenvalue)

λ is a ***dominant eigenvalue*** of a matrix A if no other eigenvalue of A has larger magnitude.

Theorem

Let A be a nonnegative matrix. If A has a positive eigenvector then the corresponding eigenvalue λ is nonnegative and dominant, i.e., $\rho(A) = \lambda$.

Proof.

See Horn & Johnson (2013) *Matrix Analysis*, Corollary 8.1.30, p. 522. □

Dominant eigenvalue of dispersal matrix M is always 1

Corollary

Consider a discrete-time metapopulation map,

$$x_i^{t+1} = \sum_{j=1}^{n} m_{ij} F(x_j^t), \quad i = 1, \ldots, n. \qquad (\heartsuit)$$

If solutions of the single patch system, $x^{t+1} = F(x^t)$, yield coherent solutions of $(\heartsuit)$ then 1 is a dominant eigenvalue of M.

Proof.

We found earlier that if solutions of the single patch map yield coherent solutions of $(\heartsuit)$ then all row sums are 1, *i.e.*, $\sum_{j=1}^{n} m_{ij} = 1$ for all i.

This is equivalent to the statement that $\mathrm{M}\boldsymbol{e} = \boldsymbol{e}$, *i.e.*, 1 is an eigenvalue of M with eigenvector $\boldsymbol{e}$.

But $\boldsymbol{e}$ is a positive vector, hence by the theorem on the previous slide, 1 is a dominant eigenvalue of M. □

Quantities that affect coherence

Maximum "reproductive rate":

- Maximum fecundity = maximum reproduction per individual per time step.
- For (single patch) logistic map, $F(x) = rx(1-x)$, maximum fecundity is r. *Note:* $r = F'(0) = \max_x |F'(x)|$.
- For any one-dimensional single species map F, we define $r = \max_x |F'(x)|$.
- More generally, single patch map can be multi-dimensional: could represent multiple species (*e.g.*, predator, prey, ...) and/or multiple states per species (*e.g.*, S, E, I, R).
- We can think of $r = \max_{\mathbf{x}} \|D_{\mathbf{x}}F\|$ as the maximum "reproductive rate" for a multi-dimensional single-patch map.
- r is relevant to coherence.

Quantities that affect coherence

Average "reproductive rate" (scalar map):

- For the scalar map $x^{t+1} = F(x^t)$, we have defined the maximum "reproductive rate" as

$$r = \max_x \left|F'(x)\right|.$$

- How should we define an *average "reproductive rate"* along a trajectory $x^0, x^1, x^2, \ldots$?
- A natural first attempt is the arithmetic mean over T steps:

$$\frac{1}{T}\sum_{t=0}^{T-1} \left|F'(x^t)\right|.$$

- To decide whether this is sensible, we need to understand how the successive local rates affect nearby trajectories.

Quantities that affect coherence

Average "reproductive rate" (scalar map):

- Compare two nearby trajectories whose densities at time t differ by δ^t.
- Their difference at the next step is

$$\begin{aligned}\delta^{t+1} &= F(x^t + \delta^t) - F(x^t) \\ &\approx F'(x^t)\,\delta^t \qquad \text{for small } \delta^t.\end{aligned}$$

- The local "reproductive rate" $|F'(x^t)|$ *multiplies* a small difference:
  - greater than 1: small differences grow over this step;
  - less than 1: small differences shrink over this step.
- A negative derivative reverses the sign of the difference.

Quantities that affect coherence

Average "reproductive rate" (scalar map):
Write F^T for *T iterations* of F, so $x^T = F^T(x^0)$.

- By the chain rule, the derivative after two steps is

$$\begin{aligned}(F^2)'(x^0) &= F'(F(x^0))F'(x^0) \\ &= F'(x^1)F'(x^0).\end{aligned}$$

- After T steps,

$$(F^T)'(x^0) = \prod_{t=0}^{T-1} F'(x^t).$$

- To first order, $\left|\delta^T\right| \approx \left|(F^T)'(x^0)\right| |\delta^0|$. Thus the successive local rates *multiply*.

Quantities that affect coherence

Average "reproductive rate" (scalar map):

- Suppose two successive local rates are 2 and $1/4$. The net factor over two steps is $2 \times (1/4) = 1/2$.
- The equivalent factor *per step* is $\sqrt{1/2} < 1$, since $\sqrt{1/2} \times \sqrt{1/2} = 1/2$. The arithmetic mean, $(2 + 1/4)/2 = 9/8 > 1$, would be misleading.
- For a trajectory of T steps, the equivalent rate per step is the *geometric mean*

$$g_T = \left[\prod_{t=0}^{T-1} |F'(x^t)| \right]^{1/T} = \left| (F^T)'(x^0) \right|^{1/T}.$$

- This gives a sensible *average "reproductive rate"* along the trajectory, accounting for how the local rates combine.

Quantities that affect coherence

Average "reproductive rate" (scalar map):

- Taking logs gives the average *logarithmic* rate:

$$\log g_T = \frac{1}{T}\sum_{t=0}^{T-1} \log\left|F'(x^t)\right| = \frac{1}{T}\log\left|(F^T)'(x^0)\right|.$$

- The average need not converge. Its upper limiting value (lim sup) defines the *Lyapunov exponent*:

$$\chi_{x^0} = \limsup_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \log\left|F'(x^t)\right|.$$

- For first-order perturbations along this trajectory:
  - $\chi_{x^0} < 0$: exponential contraction;
  - $\chi_{x^0} > 0$: perturbations can grow exponentially;
  - $\chi_{x^0} = 0$: the exponential rate alone is inconclusive.
- For a zero local rate, use $\log 0 = -\infty$.